q-HODGE FILTRATION AND $\mathrm{THH}(\mathbb{Z}_p[\zeta_p])$

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Abstract. These notes are for the talks on q-de Rham and Habiro cohomology at AMSS during 2025 fall organized by Grigory Andreychev, Shizhang Li and Zhiyou Wu. We will define q-Hodge filtration var even filtrations on solid THH. We will form the comparison map between the q-Hodge complex and $\mathrm{TC}^-(\mathrm{ku})$ and globally define the q-Hodge filtration. We will prove that our construction is indeed the q-deformation of classical Hodge filtration. Moreover, we will explain Devalapurkar's computation of THH of a special $\mathbb{E}_\infty$ algebra, which is the main gradient of our construction.

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Introduction

There are various connections between homology in cyclotomic world and de Rham cohomology for algebraic objects. The most well-known example is Hochschild-Kostant-Rosenberg theorem, which states that

$$\mathrm{H}(\mathrm{HH}_*(R)) \cong \Omega^*_{R/\mathbb{Z}}$$

where R is smooth over $\mathbb{Z}$. The later thread of this connection has witnessed a shift of focus from the spectrum to the filtration. Due to Ben Antieau [Ant19], there is an equivalence

$$\Sigma^{-2*}\mathrm{gr}^*_{mot}\mathrm{HC}^-(R) \simeq \mathrm{fil}^*_{\mathrm{Hdg}}\widehat{\mathrm{dR}}_{R/\mathbb{Z}}$$

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where the left hand side is the "double sheering" graded pieces of the motivic filtration on HC^-, and the right hand side is the Hodge filtration of complete de Rham cohomology.

Our main purpose of these lectures is to form the analogous comparison maps for q-de Rham cohomology. In order to show the relationship between TC^- and q-de Rham complex, we need to generalize the definition for Hodge filtrations to the q-case first. This will heavily rely on the computation for $\mathrm{THH}(\mathbb{Z}[\zeta_p]/\mathbb{S}[\![q-1]\!])$ finished by [Dev25] when $p > 2$, which connected this formal THH with the ku-lifting of discrete rings. Moreover, we can fit everything in the global case, but since the flaws that appear when $p = 2$. we will demand some amendment for this case.

These notes are divided as follows. The first part is devoted to make some recollections about the basic facts with regard to the synthetic Tate constructions and even filtrations on solid THH. We will also explain our requirement for the rings. In the second part, we construct the comparison map between q-de Rham complex and the 0-th graded piece of the even filtration on $\mathrm{TP}_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$ and states their relationship on filtrations. We will also explain how to fit everything in the global case and the reason that $p = 2$ case fails. In the end, we will present the proof of our main ingredient with the help of some $K(1)$-local homotopy theory Our main references are [Dev25] and [Wag].

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1. Preliminary

In this section, we recall some requirements, definitions, and facts which tither have been presented in previous talks or will be heavily demanded in the talks.

1.1. **Requirement for our rings.** We fix A to be a p-complete and p-perfectly covered δ-ring and R to be an A-algebra. Moreover, we need the following requirements for additional structures.

Assumption 1.1 (Cyclotomic assumption)**.** The ring A has a lift to a p-complete connective $\mathbb{E}_\infty$ ring $\mathbb{S}_A$ such that $\mathbb{S}_A \otimes_{\mathbb{S}_p} \mathbb{Z}_p \simeq A$. Furthermore, the Tate valued Frobenius

$$\phi_{tC_p} : \mathbb{S}_A \to \mathbb{S}_A^{tC_p}$$

agrees with the δ-ring Frobenius on A and is S^1 equivariant.

The S^1-equivariant Frobenius admits $\mathbb{S}_A$ a structure of cyclotomic base, which ensures that $\mathrm{THH}(-/\mathbb{S}_A) \simeq \mathrm{THH}(-) \otimes_{\mathrm{THH}(\mathbb{S}_A)} \mathbb{S}_A$ takes value in CycSp.

Now we state our requirement for R. We require R to be a p-complete A-algebra of bounded p^∞-tortion. We demand R to be quasi-lci over A. More over, we require that R satisfies one of the following assumptions.

Assumption 1.2 ($\mathbb{E}_1$ assumption)**.** R is p-tortion free and has a quasi-syntomic cover $R \to R_\infty$ in the sense of [[BMS19]; Definition 4.10], which satisfies that

(1) R_∞/p is relatively semiperfect over A. That means, the relative Frobenius over the δ-ring A is a surjection $R_\infty/p \otimes_{A,\phi} A \to R_\infty/p$.

(2) Denote the p-complete Čech nerve of the quasi-syntomic cover $R \to R_\infty$ by $R^\bullet_\infty$. The augmented cosimplicial diagram $R \to R^\bullet_\infty$ has a lifting to an augmented cosimplicial diagram $\mathbb{S}_R \to \mathbb{S}_{R^\bullet_\infty}$, with every term and every morphism living in $\mathrm{Alg}_{\mathbb{E}_1}(\mathrm{Mod}_{\mathbb{S}_A}(\mathrm{Sp}))$. Moreover, each term of the augmented cosimplicial diagram is p-complete and connective.

Assumption 1.3 ($\mathbb{E}_2$ assumption)**.** The ring R has a lift to a p-complete connective $\mathbb{E}_2$-algebra $\mathbb{S}_R \in \mathrm{Alg}_{\mathbb{E}_2}(\mathrm{Mod}_{\mathbb{S}_A}(\mathrm{Sp}))$ such that $\mathbb{S}_R \otimes_{\mathbb{S}_p} \mathbb{Z}_p \simeq R$.

1.2. **Synthetic fixed points and even filtrations on** THH**.** The construction will heavily rely on various even filtrations and comparisons between them. In this subsection we recall the main result of [Wag] with regard to the solid even filtrations and list some desired properties.

Recall the notation from Section 1.1. We fix an $\mathbb{E}_\infty$ ring k such that k is torsion free on even degrees. We denote $k_A := k \otimes^{\blacksquare} \mathbb{S}_A$, $k_{R^\bullet_\infty} := k \otimes^{\blacksquare} \mathbb{S}_{R^\bullet_\infty}$ and $k_R := k \otimes^{\blacksquare} \mathbb{S}_R$. In our case we usually take $k = \mathrm{ku}_p$ to be the p-complete connective K-theory. In this case we simply write ku_A, $\mathrm{ku}_{R^\bullet_\infty}$ and ku_R for k_A, $k_{R^\bullet_\infty}$ and k_R.

We need to treat the two different requirements of R with different strategies. In the situation Assumption 1.3, since our lifting of R owns an $\mathbb{E}_2$-structure, $\mathrm{THH}(k_R/k_A)$ reserves an $\mathbb{E}_1$-structure, thus we could apply Pstragowski's perfect even filtration in [Pst23] to obtain our solid even filtration. However, in the situation Assumption 1.2, we have to use a more ad hoc version of definition.

Definition 1.4 ($\mathbb{E}_1$ definition)**.** In the situation Assumption 1.2, we define

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_{\blacksquare}(k_R/k_A) := \lim_{\Delta} \tau_{\geqslant 2*}\mathrm{THH}_{\blacksquare}(k_{R^\bullet_\infty}/k_A)$$

Definition 1.5 ($\mathbb{E}_2$ definition)**.** In the situation Assumption 1.3, we define $\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_{\blacksquare}(k_R/k_A)$ to be the solid even filtration of the $\mathbb{E}_1$ ring $\mathrm{THH}_{\blacksquare}(k_R/k_A)$.

In order to define the even filtrations on $\mathrm{TC}^-_{\blacksquare}$ and $\mathrm{TP}_{\blacksquare}$, we need to apply the synthetic fixed points and synthetic Tate construction from [AR24]. We briefly introduce some basic facts about these synthetic constructions

Definition 1.6. Let $\mathbb{S}_{\mathrm{ev}} := \mathrm{fil}^*_{\mathrm{ev}}\mathbb{S} \simeq \tau_{\geqslant 2*}\mathbb{S} \in \mathrm{FilSp}$ denote the **filtered even sphere**. We define the category of **synthetic spectrum** $\mathrm{SynSp} := \mathrm{Mod}_{\mathbb{S}_{\mathrm{ev}}}(\mathrm{FilSp})$.

Definition 1.7. We define $\mathbb{T}_{\mathrm{ev}} := \mathrm{fil}^*_{\mathrm{ev}}\mathbb{S}[S^1] \in \mathrm{CAlg}(\mathrm{SynSp})$ to be the **filtered even circle**.

Lemma 1.8. *The filtered even circle $\mathbb{T}_{\mathrm{ev}}$ is dualizable in* SynSp*.*

We recall that a map of $\mathbb{E}_\infty$-rings $A \to B$ is **even faithfully flat** (eff) if for every $\mathbb{E}_\infty$-ring map $A \to C$ with C even, $C \otimes_A B$ is even and $\pi_*(C \otimes_A B)$ is faithfully flat over $\pi_*(C)$ in every degree. We say that A is **eff-locally even** if it admits an eff map $A \to B$ with B even.

Proposition 1.9. *If B is an eff-locally even $\mathbb{E}_\infty$ ring, then $\mathbb{T}_{\mathrm{ev}} \otimes_{\mathbb{S}_{\mathrm{ev}}} \mathrm{fil}^*_{\mathrm{ev}}B \to \mathrm{fil}^*_{\mathrm{ev}}(B[S^1])$ is an equivalence.*

Proof. Let $B \to C$ be an eff cover with C even. By definition of eff $B[S^1] \to C[S^1]$ is also an eff cover and $C^{\otimes_B n}$ is even for every n. Therefore, we have

$$\mathrm{fil}^*_{\mathrm{ev}}(C^{\otimes_B n}[S^1]) \simeq \tau_{\geqslant 2*-1}(C^{\otimes_B n}[S^1])$$

Applying eff descent in [HRW22], we obtain

$$\begin{aligned}\mathrm{fil}^*_{\mathrm{ev}}(B[S^1]) \simeq & \lim_{\Delta} \tau_{\geqslant 2*-1}(C^{\otimes_B n}[S^1]) \\ \simeq & \lim_{\Delta} \left(\tau_{\geqslant 2*} C^{\otimes_B n} \oplus \tau_{\geqslant 2*-2}(C^{\otimes_B n})[1]\right) \\ \simeq & \lim_{\Delta}(\tau_{\geqslant 2*} C^{\otimes_B n}) \oplus \lim_{\Delta}(\tau_{\geqslant 2*-2} C^{\otimes_B n})[1] \\ \simeq & \mathrm{fil}^{*-1}_{\mathrm{ev}}(B)[1] \oplus \mathrm{fil}^*_{\mathrm{ev}}(B) \\ \simeq & \mathbb{T}_{\mathrm{ev}} \otimes_{\mathbb{S}_{\mathrm{ev}}} \mathrm{fil}^*_{\mathrm{ev}} B\end{aligned}$$

□

Let $S^1 \to S^1$ be multiplication by n. This map induces an $\mathbb{E}_\infty$-map $\mathbb{S}[S^1] \to \mathbb{S}[S^1]$, hence induces a map on filtered objects $\rho(n) : \mathbb{T}_{\mathrm{ev}} \to \mathbb{T}_{\mathrm{ev}}$. We use $\rho(n)_* \mathbb{T}_{\mathrm{ev}}$ to denote $\mathbb{T}_{\mathrm{ev}}$ regarded as a $\mathbb{T}_{\mathrm{ev}}$-module via $\rho(n)$.

Lemma 1.10. *For B an even $\mathbb{E}_\infty$-ring with trivial action, we have a standard equivalence of $\mathrm{fil}^*_{\mathrm{ev}} B[S^1]$-modules*

$$\mathrm{fil}^*_{\mathrm{ev}} B \otimes_{\mathbb{S}_{\mathrm{ev}}} \rho(n)_* \mathbb{T}_{\mathrm{ev}} \to \mathrm{fil}^*_{\mathrm{ev}}(B[S^1/n])$$

Proof. This follows analogously from Proposition 1.9. □

Definition 1.11. For $M \in \mathrm{Mod}_{\mathbb{T}_{\mathrm{ev}}}(\mathrm{SynSp})$, we define the **synthetic orbit** and **synthetic fixed points** as

$$M_{\mathbb{T}_{\mathrm{ev}}} := \mathbb{S}_{\mathrm{ev}} \otimes_{\mathbb{T}_{\mathrm{ev}}} M \quad M^{\mathbb{T}_{\mathrm{ev}}} := \mathrm{map}_{\mathbb{T}_{\mathrm{ev}}}(\mathbb{S}_{\mathrm{ev}}, M)$$

They are respectively left and right adjoint of the restriction of scalars $\mathrm{SynSp} \to \mathrm{SynSp}_{\mathbb{T}_{\mathrm{ev}}}$.

We now construct the synthetic Tate. Note that there are double degrees on any synthetic spectra $\mathbb{S}_{\mathrm{ev}}$. We use the parenthesis to denote shifting on filtered degree and square brackets to denote shifting on spectrum degree.

Construction 1.12 (Synthetic S^1-Tate)**.** By dualizability of $\mathbb{T}_{\mathrm{ev}}$, we have an equivalence $\alpha : \mathbb{T}^\vee_{\mathrm{ev}} \to \mathbb{T}_{\mathrm{ev}}(-1)[-1] \simeq \mathbb{T}_{\mathrm{ev}} \otimes_{\mathbb{S}_{\mathrm{ev}}} \mathbb{S}_{\mathrm{ev}}(-1)[-1]$

Suppose we are given $M \in \mathrm{Mod}_{\mathbb{T}_{\mathrm{ev}}}(\mathrm{SynSp})$, we have a map

$$\begin{aligned}M_{\mathbb{T}_{\mathrm{ev}}}(1)[1] := \mathbb{S}_{\mathrm{ev}}(1)[1] \otimes_{\mathbb{S}_{\mathrm{ev}}} M_{\mathbb{T}_{\mathrm{ev}}} &= \mathbb{S}_{\mathrm{ev}}(1)[1] \otimes_{\mathbb{S}_{\mathrm{ev}}} M \otimes_{\mathbb{T}_{\mathrm{ev}}} \mathbb{S}_{\mathrm{ev}} \\ &\to \mathbb{S}_{\mathrm{ev}}(1)[1] \otimes_{\mathbb{S}_{\mathrm{ev}}} M \otimes_{\mathbb{T}_{\mathrm{ev}}} \mathbb{T}^\vee_{\mathrm{ev}} \\ &\simeq^{\alpha} \mathbb{S}_{\mathrm{ev}}(1)[1] \otimes_{\mathbb{S}_{\mathrm{ev}}} M \otimes_{\mathbb{T}_{\mathrm{ev}}} \mathbb{T}_{\mathrm{ev}} \otimes_{\mathbb{S}_{\mathrm{ev}}} \mathbb{S}_{\mathrm{ev}}(-1)[-1] \\ &\simeq M\end{aligned}$$

The left hand side is endowed with trivial $\mathbb{T}_{\mathrm{ev}}$-action so the map factors through $M^{\mathbb{T}_{\mathrm{ev}}}$. We denote the map $M_{\mathbb{T}_{\mathrm{ev}}}(1)[1] \to M^{\mathbb{T}_{\mathrm{ev}}}$ by $\mathrm{Nm}_{\mathbb{T}_{\mathrm{ev}}}$ and define $M^{t\mathbb{T}_{\mathrm{ev}}}$ to be the cofiber of $\mathrm{Nm}_{\mathbb{T}_{\mathrm{ev}}}$.

Similarly, we can consider the $\mathbb{Z}/n$ fixed points, orbits and Tate constructions.

Definition 1.13. Given $M \in \mathrm{SynSp}_{\mathbb{T}_{\mathrm{ev}}}$, we define the $\mathbb{Z}/n$-orbits and $\mathbb{Z}/n$-fixed points to be

$$M_{\mathbb{Z}/n,\mathrm{ev}} := \rho(n)_* \mathbb{T}_{\mathrm{ev}} \otimes_{\mathbb{T}_{\mathrm{ev}}} M \quad M^{\mathbb{Z}/n,\mathrm{ev}} := \mathrm{map}_{\mathbb{T}_{\mathrm{ev}}}(\rho(n)_* \mathbb{T}_{\mathrm{ev}}, M)$$

Construction 1.14 (Synthetic $\mathbb{Z}/n$-Tate)**.** By symmetric monoidality of $\rho(n)_*$, we have

$$\begin{aligned} M_{\mathbb{Z}/n,\mathrm{ev}} &= \rho(n)_*\mathbb{T}_{\mathrm{ev}} \otimes_{\mathbb{T}_{\mathrm{ev}}} M \\ &\simeq \mathbb{S}_{\mathrm{ev}}(1)[1] \otimes \rho(n)_*\mathbb{T}_{\mathrm{ev}}^{\vee} \otimes_{\mathbb{T}_{\mathrm{ev}}} M \\ &\to \mathbb{S}_{\mathrm{ev}}(1)[1] \otimes \mathbb{T}_{\mathrm{ev}}^{\vee} \otimes_{\mathbb{T}_{\mathrm{ev}}} M \\ &\simeq \mathbb{S}_{\mathrm{ev}}(1)[1] \otimes \mathbb{S}_{\mathrm{ev}}(-1)[-1] \otimes \mathbb{T}_{\mathrm{ev}} \otimes_{\mathbb{T}_{\mathrm{ev}}} M \\ &\simeq M \end{aligned}$$

Similar argument shows that the map factors through $M^{\mathbb{Z}/n,\mathrm{ev}}$. We denote this map by $\mathrm{Nm}_n : M_{\mathbb{Z}/n,\mathrm{ev}} \to M^{\mathbb{Z}/n,\mathrm{ev}}$, and define the cofiber of this map to be $M^{t\mathbb{Z}/n,\mathrm{ev}}$.

Remark 1.15. Following the method of [NS18], one can show that both $(-)^{t\mathbb{T}_{\mathrm{ev}}}$ and $(-)^{t\mathbb{Z}/n,\mathrm{ev}}$ are lax symmetric monoidal, and the natural transformation $(-)^{t\mathbb{T}_{\mathrm{ev}}} \to (-)^{t\mathbb{Z}/n,\mathrm{ev}}$ is lax symmetric monoidal as well.

It is natural to ask how these synthetic constructions: orbits, fixed points and Tate behave under even filtration functors. Fortunately, the situation is good if we consider only the Hahn-Raksit-Wilson even filtration.

Theorem 1.16 ([AR24]; Theorem 2.75)**.** *Let M be an $\mathbb{E}_\infty$-ring spectrum such that $\mathrm{fil}^*_{\mathrm{ev}}M$ is a $\mathbb{T}_{\mathrm{ev}}$-module. Then by symmetric monoidality of colimit functor, M is naturally equipped with a $\mathbb{T}$-action. Fix $n \geqslant 1$.*

1. *The natural maps $\mathrm{colim}_{-\infty}(\mathrm{fil}^*_{\mathrm{ev}}M)_{\mathbb{T}_{\mathrm{ev}}} \to M_{hS^1}$ and $\mathrm{colim}_{-\infty}(\mathrm{fil}^*_{\mathrm{ev}}M)_{\mathbb{Z}/n,\mathrm{ev}} \to M_{h\mathbb{Z}/n}$ are equivalences.*
2. *If $\mathrm{fil}^*_{\mathrm{ev}}M$ is complete, then so are $(\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{T}_{\mathrm{ev}}}$ and $(\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{Z}/n,\mathrm{ev}}$.*
3. *Suppose that $\mathrm{fil}^i_{\mathrm{ev}}M \to M$ is $2i$-truncated, then*

$$\left((\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{T}_{\mathrm{ev}}}\right)^i \to M^{hS^1}$$

and

$$\left((\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{Z}/n,\mathrm{ev}}\right)^i \to M^{h\mathbb{Z}/n}$$

are also $2i$-truncated.

In particular, the underlying spectra of $(\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{T}_{\mathrm{ev}}}$ and $(\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{Z}/n,\mathrm{ev}}$ are exactly M^{hS^1} and $M^{h\mathbb{Z}/n}$.

4. *In the situation of (3), if $\mathrm{fil}^*_{\mathrm{ev}}M \simeq \tau_{\geqslant 2*}M$, then $(\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{T}_{\mathrm{ev}}}$ is the double speed Whitehead tower of M^{hS^1}. If additionally M is even, then $(\mathrm{fil}^*_{\mathrm{ev}}M)^{\mathbb{Z}/n,\mathrm{ev}}$ is the double speed Whitehead tower of $M^{h\mathbb{Z}/n}$.*

We can now construct the desired even filtrations on $\mathrm{TC}^-_{\blacksquare}$ and $\mathrm{TP}_{\blacksquare}$.

Definition 1.17. In both of the situations, we define the even filtration on $\mathrm{TC}^-_{\blacksquare}$ and $\mathrm{TP}_{\blacksquare}$ to be

$$\begin{aligned} \mathrm{fil}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(k_R/k_A) &:= \left(\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_{\blacksquare}(k_R/k_A)\right)^{h\mathbb{T}_{\mathrm{ev}}} \\ \mathrm{fil}^*_{\mathrm{ev}}\mathrm{TP}_{\blacksquare}(k_R/k_A) &:= \left(\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_{\blacksquare}(k_R/k_A)\right)^{t\mathbb{T}_{\mathrm{ev}}} \end{aligned}$$

In both Assumption 1.2 and Assumption 1.3, our even filtrations on $\mathrm{THH}_{\blacksquare}$, $\mathrm{TC}^-_{\blacksquare}$ and $\mathrm{TP}_{\blacksquare}$ own cosimplicial resolutions, which will become useful for us to check various properties in practice.

Proposition 1.18. *In the situation Assumption 1.3, let $P = \mathbb{Z}[x_i|i \in I]$ be a polynomial ring such that $P \to R$ is surjective. Let $\mathbb{S}_P = \mathbb{S}[x_i|i \in I]$. There is an $\mathbb{E}_1$-map $\mathbb{S}_P \to k_R$ which can upgrade to an $\mathbb{E}_2$-map. Denote the p-complete Čech nerve of $\mathbb{S} \to \mathbb{S}_P$ by $\mathbb{S} \to \mathbb{S}_{\widehat{P}_p^\bullet}$. Then we obtain an augmented cosimplicial diagram*

$$\mathrm{THH}_\blacksquare(k_R/k_A) \longrightarrow \mathrm{THH}_\blacksquare(k_R/k_A \otimes^\blacksquare \mathbb{S}_{\widehat{P}_p^\bullet})$$

The cosimplicial resolution induces a canonical equivalence

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_\blacksquare(k_R/k_A) \longrightarrow \lim_\Delta \tau_{\geqslant 2*}\mathrm{THH}_\blacksquare(k_R/k_A \otimes^\blacksquare \mathbb{S}_{\widehat{P}_p^\bullet})$$

Therefore, in both Assumption 1.2 and Assumption 1.3, $\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_\blacksquare(k_R/k_A)$ admits a cosimplicial resolution.

Corollary 1.19. *In the situation Assumption 1.2, the cosimplicial resolution induces equivalences*

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{TC}^-_\blacksquare(k_R/k_A) \xrightarrow{\simeq} \lim_\Delta \tau_{\geqslant 2*}\mathrm{TC}^-_\blacksquare(k_{R^\bullet_\infty}/k_A)$$

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{TP}_\blacksquare(k_R/k_A) \xrightarrow{\simeq} \lim_\Delta \tau_{\geqslant 2*}\mathrm{TP}_\blacksquare(k_{R^\bullet_\infty}/k_A)$$

In the situation Assumption 1.3, the cosimplicial resolution induces equivalences

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{TC}^-_\blacksquare(k_R/k_A) \xrightarrow{\simeq} \lim_\Delta \tau_{\geqslant 2*}\mathrm{TC}^-_\blacksquare(k_R/k_A \otimes^\blacksquare \mathbb{S}_{\widehat{P}_p^\bullet})$$

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{TP}_\blacksquare(k_R/k_A) \xrightarrow{\simeq} \lim_\Delta \tau_{\geqslant 2*}\mathrm{TP}_\blacksquare(k_R/k_A \otimes^\blacksquare \mathbb{S}_{\widehat{P}_p^\bullet})$$

We conclude this subsection by comparing our even filtrations with the Hahn-Raksit-Wilson even filtrations, which follows by comparing corresponding cosimplicial resolution.

Proposition 1.20. *Suppose $k = \mathbb{Z}$, A and R are as in Section 1.1. Then the solid even filtrations defined above*

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{HH}_\blacksquare(R/A) \qquad \mathrm{fil}^*_{\mathrm{ev}}\mathrm{HC}^-_\blacksquare(R/A) \qquad \mathrm{fil}^*_{\mathrm{ev}}\mathrm{HP}_\blacksquare(R/A)$$

agrees with the Hahn-Raksit-Wilson even filtrations

$$\mathrm{fil}^*_{\mathrm{HRW}}\mathrm{HH}_\blacksquare(R/A) \qquad \mathrm{fil}^*_{\mathrm{HRW}}\mathrm{HC}^-_\blacksquare(R/A) \qquad \mathrm{fil}^*_{\mathrm{HRW}}\mathrm{HP}_\blacksquare(R/A)$$

1.3. **∞-group of units.** In this section, we develop tools that will be needed in Section 4.

Definition 1.21. Let R be an $\mathbb{E}_1$- ring. We define the **group of units** of R as the pullback in the category of spaces.

$$\begin{array}{ccc} \mathrm{GL}_1(R) & \longrightarrow & \Omega^\infty R \\ \downarrow & \lrcorner & \downarrow \\ \pi_0(\Omega^\infty R)^\times & \longrightarrow & \pi_0(\Omega^\infty R) \end{array}$$

when R is an $\mathbb{E}_\infty$-ring, $\mathrm{GL}_1(R)$ becomes an $\mathbb{E}_\infty$-space, which can be identified with a connective spectrum $\mathfrak{gl}_1(R)$. We call $\mathfrak{gl}_1(-)$ the ∞-**group of units** functor.

The functor $\mathfrak{gl}_1(-)$ admits a left adjoint, which is the ∞-**group**-∞-**ring** functor $\mathbb{S}[-]$.

Example 1.22 (Snaith's theorem)**.** There are equivalences of $\mathbb{E}_\infty$ rings

$$\mathrm{KU} \simeq \mathbb{S}[\mathrm{BU}(1)][\beta^{-1}]$$

We also have the following description for homotopy groups of ∞-group of units.

Proposition 1.23. *The homotopy group of $\mathfrak{gl}_1(R)$ is*

$$\pi_n(\mathfrak{gl}_1(R)) = \begin{cases} \pi_0(R)^\times & \text{for } n = 0 \\ \pi_n(R) & \text{otherwise} \end{cases}$$

We will need the following equivalence which is induced by Bousfield-Kuhn functor, which is defined in [Kuh06].

Theorem 1.24. *Let R be an $\mathbb{E}_\infty$-ring, then there is an equivalence*

$$L_{K(1)}\mathfrak{gl}_1(R) \simeq L_{K(1)}R$$

2. Define the q-Hodge filtration

In this section, we will construct our desired Hodge filtration on q-de Rham complex. We first state our main ingredient as follows, which will be proven in the next section.

Notations 2.1. We define $j_{p,0}$ to be the $\mathbb{E}_\infty$-ring $\tau_{\geqslant 0}(\mathrm{KU}_p^{h\mathbb{Z}})$ with $\mathbb{Z} \subseteq \mathbb{Z}_p$ generated by $1+p$. We use $\mathbb{S}[\![q^{1/p}-1]\!]$ to denote $\mathbb{S}[\![q^{1/p}-1]\!] \simeq \lim \mathbb{S}[q^{1/p}]/(q^\alpha - 1) \simeq \lim \mathbb{S}[\mathbb{Z}/p^\alpha]$. The second equivalence is given by [[Lur18]; Proposition 2.2.12]. Thus there is a standard $\mathbb{Z}_p^\times$ action on $\mathbb{S}[\![q^{1/p}-1]\!]$.

Theorem 2.2. *There is a commutative diagram of $\mathbb{E}_\infty$ rings for $p > 2$.*

$$\begin{array}{ccccc}
j_{p,0} & \longrightarrow & \tau_{\geqslant 0} j_{p,0}^{t\mathbb{Z}/p} & \xrightarrow{\simeq} & \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \\
\downarrow & \ulcorner & \downarrow & & \downarrow \\
\mathrm{ku}_p & \longrightarrow & \tau_{\geqslant 0}\mathrm{ku}_p^{t\mathbb{Z}/p} & \xrightarrow{\simeq} & \mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) \\
\downarrow & & \downarrow & & \downarrow \\
\mathbb{Z}_p & \longrightarrow & \tau_{\geqslant 0}\mathbb{Z}_p^{t\mathbb{Z}/p} & \xrightarrow{\simeq} & \mathrm{THH}(\mathbb{F}_p)
\end{array}$$

Moreover, the pushout equivalence

$$\mathrm{ku}_p \otimes_{j_{p,0}} \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \simeq \tau_{\geqslant 0}\mathrm{ku}_p \simeq \mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!])$$

is an equivalence of $S^1 \times \mathbb{Z}_p$-equivariant $\mathbb{E}_\infty$-$\mathbb{S}[\![q^{1/p}-1]\!]$-algebra, where $\mathbb{S}[\![q^{1/p}-1]\!]$ acts on $\mathbb{Z}_p[\zeta_p]$ via the map $q^{1/p} \mapsto \zeta_p$.

Remark 2.3. The diagram is in fact a commutative diagram of $\mathbb{E}_\infty$-cyclotomic rings. There is a standard cyclotomic structure on $\tau_{\geqslant 0}(X^{t\mathbb{Z}/p})$ for any connective cyclotomic spectrum X, where the new cyclotomic structure is given by applying the $\mathbb{Z}/p$-Tate on $X \simeq \tau_{\geqslant 0}X \to \tau_{\geqslant 0}(X^{t\mathbb{Z}/p})$.

Remark 2.4. We need to explain further what the map $\mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) \to \mathrm{THH}(\mathbb{F}_p)$ is. To see this, we need to regard both sides as $\mathbb{Z}_p$-algebras, by Devalapurkar's computation [Dev25], we have $\pi_*(\tau_{\geqslant 0}\mathrm{ku}_p^{t\mathbb{Z}/p}) \simeq \mathbb{Z}_p[\zeta_p][u]$ with a single generater u in degree 2. The second vertical arrow in the middle sends ζ_p to 1. Therefore, the second vertical arrow $\mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) \to \mathrm{THH}(\mathbb{F}_p)$ is given by modulo $q-1$.

We are now going to construct our comparison map. Let A and R be defined as in Section 1.1. We consider the S^1-equivariant map of $\mathbb{E}_\infty$ rings

$$\mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) \simeq \tau_{\geqslant 0}\mathrm{ku}_p^{t\mathbb{Z}/p} \to \mathrm{ku}_p^{t\mathbb{Z}/p}$$

We reform everything in the solid world. After tensoring with the relative Frobenius

$$\mathrm{THH}_\blacksquare(\mathbb{S}_R/\mathbb{S}_A) \otimes^\blacksquare_{\mathbb{S}_A,\phi_{tC_p}} \mathbb{S}_A \to \mathrm{THH}_\blacksquare(\mathbb{S}_R/\mathbb{S}_A)^{t\mathbb{Z}/p}$$

we obtain the following composition square.

$$\begin{array}{ccc} \mathrm{THH}_\blacksquare(\mathbb{S}_R/\mathbb{S}_A) \otimes^\blacksquare_{\mathbb{S}_A,\phi_{tC_p}} \mathbb{S}_A \otimes^\blacksquare \mathrm{THH}_\blacksquare(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) & \longrightarrow & \mathrm{THH}_\blacksquare(\mathbb{S}_R/\mathbb{S}_A)^{t\mathbb{Z}/p} \otimes^\blacksquare \mathrm{ku}_p^{t\mathbb{Z}/p} \\ \simeq\downarrow & & \downarrow \\ \mathrm{THH}_\blacksquare\big((R \otimes^\blacksquare_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big) & \dashrightarrow & \mathrm{THH}^\blacksquare(\mathrm{ku}_R/\mathrm{ku}_A)^{t\mathbb{Z}/p} \end{array}$$

The left equivalence is because THH is a symmetric monoidal functor and the right vertical arrow is given by the lax symmetric monoidality of $(-)^{t\mathbb{Z}/p}$. Here we use our cyclotomic assumption for A.

Proposition 2.5. *The obtained map*

$$\mathrm{THH}_\blacksquare\big((R \otimes^\blacksquare_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big) \to \mathrm{THH}^\blacksquare(\mathrm{ku}_R/\mathrm{ku}_A)^{t\mathbb{Z}/p}$$

sends the generater $u \in \pi_2$ to a unit. Thus the above map induces a map

$$\mathrm{THH}_\blacksquare\big((R \otimes^\blacksquare_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big)[u^{-1}] \to \mathrm{THH}^\blacksquare(\mathrm{ku}_R/\mathrm{ku}_A)^{t\mathbb{Z}/p}$$

Proof. We show that the map

$$\mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) \to \mathrm{ku}_p^{t\mathbb{Z}/p}$$

sends the generator u to a unit. This could be checked after modulo $q-1$. Therefore, it suffices to show that the map

$$\mathrm{THH}(\mathbb{F}_p) \to \mathbb{Z}_p^{t\mathbb{Z}/p}$$

sends the generator $u \in \pi_2(\mathrm{THH}(\mathbb{F}_\mathrm{p}))$ to a unit. This is obvious, since by [NS18], $\mathbb{F}_p[u] \simeq \mathrm{THH}(\mathbb{F}_p)$ is identified with the connective part of $\mathbb{Z}_p^{t\mathbb{Z}/p} \simeq \mathbb{F}_p[\pm u]$. □

The map from the above proposition is S^1-equivariant. After applying homotopy fixed points, we obtain a base map

$$\psi_R : \mathrm{TC}^-_{\blacksquare}\big((R \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big)[u^{-1}]^{\wedge}_{(p,q-1)} \to \mathrm{TP}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

We now construct our desired comparision map on filtration levels. Recall that we have required our ring R to satisfy either of the two different prerequisites Assumption 1.2 and Assumption 1.3. Since these two requirements admit the liftings $\mathbb{S}_R$ and $\mathbb{S}_A$ with different monoidalities, we need to treat them with different strategies.

Construction 2.6 ($\mathbb{E}_1$ construction)**.** In the situation Assumption 1.2, we define the filtered map to be

$$\psi^*_R : \lim_{\Delta} \mathrm{TC}^-_{\blacksquare}\big((R^{\bullet}_{\infty} \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big)[u^{-1}]^{\wedge}_{(p,q-1)} \to \lim_{\Delta} \mathrm{TP}^{\blacksquare}(\mathrm{ku}_{R^{\bullet}_{\infty}}/\mathrm{ku}_A)$$

which is induced by the limit of $(\psi_R)^{hS^1}$. By quasi-syntomic descent and Corollary 1.19, the left side agrees with the Bhatt-Morrow-Scholze filtration and the right side agree with our even filtration.

Construction 2.7 ($\mathbb{E}_2$ construction)**.** In the situation Assumption 1.3, we obtain a map of $\mathbb{E}_1$-rings

$$\mathrm{THH}_{\blacksquare}\big((R \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big)[u^{-1}] \to \mathrm{THH}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)^{t\mathbb{Z}/p}$$

Thus we could apply the usual HKR even filtration and then apply $(-)^{h\mathbb{T}_{\mathrm{ev}}}$ in the sense of [AR24].

$$\Big(\mathrm{fil}^*_{\mathrm{ev}}\big(\mathrm{THH}_{\blacksquare}\big((R \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big)[u^{-1}]\big)\Big)^{h\mathbb{T}_{\mathrm{ev}}} \to \Big(\mathrm{fil}^*_{\mathrm{ev}}\big(\mathrm{THH}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)^{t\mathbb{Z}/p}\big)\Big)^{h\mathbb{T}_{\mathrm{ev}}}$$

By [[Wag],Corollary 6.23], the left side is equavalent to the desired even filtration. On the right we could take the homotopy fixed points out of the filtration. Thus we got a composition of maps

$$\begin{aligned}
\mathrm{fil}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}\big((R \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big)[u^{-1}]^{\wedge}_{(p,q-1)} &\simeq \Big(\mathrm{fil}^*_{\mathrm{ev}}\big(\mathrm{THH}_{\blacksquare}\big((R \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big)[u^{-1}]\big)\Big)^{h\mathbb{T}_{\mathrm{ev}}} \\
&\to \Big(\mathrm{fil}^*_{\mathrm{ev}}\big(\mathrm{THH}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)^{t\mathbb{Z}/p}\big)\Big)^{h\mathbb{T}_{\mathrm{ev}}} \\
&\to \Big(\mathrm{fil}^*_{\mathrm{ev}}\big(\mathrm{THH}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)\big)^{t\mathbb{Z}/p_{\mathrm{ev}}}\Big)^{h\mathbb{T}_{\mathrm{ev}}} \\
&\simeq \mathrm{fil}^*_{\mathrm{ev}}\mathrm{TP}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)
\end{aligned}$$

It remains to prove the last equivalence. To see this, first observe that by [[Wag], Corollary 6.16.], the synthetic Tate construction $(-)^{t\mathbb{Z}/p}$ commutes with cosimplicial limits by some connectivity arguments. Thus by the cosimplicial resolution from Proposition 1.18, we may reduce to the case that $\mathrm{THH}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$ is already even. The equivalence follows from Theorem 1.16 and the classical result that $(-)^{tS^1} \simeq ((-)^{t\mathbb{Z}/p})^{hS^1/p}$.

Remark 2.8. We need to make it more explicit about the second arrow in the above equation. In general, let T be a complex orientable even $\mathbb{E}_\infty$-ring and for any S^1-equivariant $\mathbb{E}_1$-T-algebra M with $M^{h\mathbb{Z}/p}$ homologically even over $T^{h\mathbb{Z}/p}$, then we have

$$\begin{aligned}\mathrm{fil}^*_{\mathrm{ev}/T^{t\mathbb{Z}/p}} M^{t\mathbb{Z}/p} &\simeq \mathrm{fil}^*_{\mathrm{ev}} T^{tS^1} \otimes^{\blacksquare}_{\mathrm{fil}^*_{\mathrm{ev}} T^{hS^1}} \mathrm{fil}^*_{\mathrm{ev}/T^{h\mathbb{Z}/p}} M^{h\mathbb{Z}/p} \\ &\to \mathrm{fil}^*_{\mathrm{ev}} T^{tS^1} \otimes^{\blacksquare}_{\mathrm{fil}^*_{\mathrm{ev}} T^{hS^1}} \left(\mathrm{fil}^*_{\mathrm{ev}/T^{h\mathbb{Z}/p}} M\right)^{h\mathbb{Z}/p_{\mathrm{ev}}} \\ &\to \left(\mathrm{fil}^*_{\mathrm{ev}/T^{t\mathbb{Z}/p}} M\right)^{t\mathbb{Z}/p_{\mathrm{ev}}}\end{aligned}$$

The comparison map is induced from the square

$$\begin{array}{ccc} (-)^{h\mathbb{T}} & \longrightarrow & (-)^{h\mathbb{Z}/p} \\ \downarrow & & \downarrow \\ (-)^{t\mathbb{T}} & \longrightarrow & (-)^{t\mathbb{Z}/p} \end{array}$$

which comes from [[NS18], Corollary I.4.3] and we use both the classical form and the synthetic form. And the first isomorphism is obtained by checking graded pieces, or in our specific case, follows from [[Wag] Corollary 6.19].

Now for both of the situations we have constructed a filtered comparison map

$$\psi^*_R : \mathrm{fil}^*_{\mathrm{ev}} \mathrm{TC}^-_{\blacksquare}\left((R \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\right)[u^{-1}]^{\wedge}_{(p,q-1)} \to \mathrm{fil}^*_{\mathrm{ev}} \mathrm{TP}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

We use this comparison map to construct q-Hodge filtrations. We first state the relation between q-de rham cohomology and the left side of our filtered map.

Theorem 2.9 ([Wag], Proposition A.17 ; [BS22], Theorem 16.18)**.** *There is an equivalence*

$$q\text{-}\mathrm{dR}_{R/A} \simeq \mathrm{gr}^0_{\mathrm{HRW}} \mathrm{TC}^-_{\blacksquare}\left((R \otimes^{\blacksquare}_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\right)[u^{-1}]^{\wedge}_{(p,q-1)}$$

Therefore we can apply ψ^0_R. Notice that on the non positive graded pieces, we have

$$\mathrm{gr}^0_{\mathrm{ev}} \mathrm{TP}^{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A) \simeq \mathrm{gr}^0_{\mathrm{ev}} \mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

Therefore we obtain a comparison map

$$\psi^0_R : q\text{-}\mathrm{dR}_{R/A} \to \mathrm{gr}^0_{\mathrm{ev}} \mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

Definition 2.10. For a stable infinity category $\mathcal{C}$, we use $\mathrm{Gr}(\mathcal{C})$ denote the graded object of $\mathcal{C}$. Then we define the **double shearing** functor to be

$$\begin{aligned}\Sigma^{2*} : \mathrm{Gr}(\mathcal{C}) &\to \mathrm{Gr}(\mathcal{C}) \\ \{\mathrm{gr}^n(X)\} &\mapsto \{\mathrm{gr}^n(X)[2n]\}\end{aligned}$$

Remark 2.11. The double shearing functor is $\mathbb{E}_1$-monoidal. Indeed, one can construct these functors as in [Rak20]. The symmetric monoidal structure on $\mathrm{Gr}(\mathcal{C})$ is defined var Day convolution, then the $\mathbb{E}_1$-monoidality can be checked on the corresponding functor category. In fact, it is even $\mathbb{E}_2$-monoidal by [Dev+23], but we won't use this fact.

Construction 2.12. We construct the **q-Hodge filtration** to be the pullback of the graded $\mathbb{Z}_p[\beta][\![t]\!] \simeq (q-1)^*\mathbb{Z}_p[\![q-1]\!]$-modules

$$\begin{array}{ccc} \mathrm{fil}^*_{\mathrm{qHdg}} q-\mathrm{dR}_{R/A} & \longrightarrow & \Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A) \\ \downarrow & & \downarrow \\ q-\mathrm{dR}_{R/A} & \longrightarrow & \mathrm{gr}^0_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A) \end{array}$$

Remark 2.13. To see that $\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$ is indeed a $\mathbb{Z}_p[\beta][\![t]\!]$-module, we first note that $\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$ is a graded module over $\mathrm{gr}^*_{\mathrm{ev}}(\mathrm{ku}^{hS^1}) \simeq \Sigma^{2*}\pi_{2*}(\mathrm{ku}^{hS^1})$. Therefore after the double shearing, the filtration $\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$ becomes a $\pi_{2*}(\mathrm{ku}^{hS^1}) \simeq \mathbb{Z}_p[\beta][\![t]\!]$-module.

3. Proof of the main theorem

In this section we sketch the proof of our main theorem, which implies that our construction of q-Hodge filtration is indeed a q-deformation of the usual Hodge filtration. Our main theorem is as follows

Theorem 3.1. *We let $p > 2$ be a fixed prime. Let A and R satisfy the requirements from Section 1.1.*

(1) The map from Construction 2.12 induces an equivalence

$$\mathrm{fil}^*_{\mathrm{qHdg}} q-\widehat{\mathrm{dR}}_{R/A} \xrightarrow{\simeq} \Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

(2) After modulo $\beta = q-1$ in degree 1, we got the equivalence

$$\mathrm{fil}^*_{\mathrm{qHdg}} q-\mathrm{dR}_{R/A} \otimes_{\mathbb{Z}_p[\beta][\![t]\!]} \mathbb{Z}_p[\beta] \xrightarrow{\simeq} \mathrm{fil}^*_{\mathrm{Hdg}}\mathrm{dR}_{R/A}$$

(3) We also have an equivalence

$$\mathrm{fil}^*_{\mathrm{qHdg}} q-\mathrm{dR}_{R/A}[p^{-1}]^{\wedge}_{q-1} \xrightarrow{\simeq} \mathrm{fil}^*_{\mathrm{Hdg},q-1}\mathrm{dR}_{R/A}[p^{-1}][\![q-1]\!]$$

3.1. **Proof of the main theorem.** We first show that no matter what situation of R we choose, the comparison map dosen't rely on the spherical lift.

Lemma 3.2. *After modulo $q-1$, the comparison map*

$$\psi^0_R : q-\mathrm{dR}_{R/A} \to \mathrm{gr}^0_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

doesn't rely on the choice of the lifting $\mathbb{S}_R$.

In order to see this, we do everything modulo β. We need to give a new construction of the comparison map after modulo β.

We first need some computation of [[DR25]; Theorem 0.1.4]

Theorem 3.3 ([DR25])**.** *Let $j_p := \tau_{\geqslant 0}\mathbb{S}_{K(1)}$ be the connective cover of $K(1)$-local sphere.*

(1) *There exists an equivalence* $\tau_{\geqslant 0}(j_p^{t\mathbb{Z}/p}) \simeq \mathrm{THH}(\mathbb{Z}_p)$*, which fits into a commutative diagram*

$$\begin{array}{ccc} j_p & \longrightarrow & \mathrm{THH}(\mathbb{Z}_p) \\ \downarrow & & \downarrow \\ \mathbb{Z}_p & \longrightarrow & \mathrm{THH}(\mathbb{F}_p) \end{array}$$

(2) *Both of the horizontal maps are* S^1*-nilpotent, that means for any* S^1*-equivariant spectrum* X*, the maps* $X \otimes j_p \to X \otimes \mathrm{THH}(\mathbb{Z}_p)^\wedge_p$ *and* $X \otimes \mathbb{Z}_p \to X \otimes \mathrm{THH}(\mathbb{F}_p)$ *become equivalence after applying* $(-)^{tS^1}$*.*

Proof of Lemma 3.2. We check everything by modulo $q-1$. In this case, the corresponding map

$$\overline{\psi}_R : \mathrm{TC}^-_\blacksquare\big((R \otimes^\blacksquare_{\mathbb{Z}_p} \mathbb{F}_p \otimes^\blacksquare_{A,\phi} A)/\mathbb{S}_A\big)[u^{-1}]^\wedge_p \to \mathrm{HP}^\blacksquare(R/A)$$

is formed analogously by considering the following composition diagram

$$\begin{array}{ccc} \mathrm{THH}_\blacksquare(\mathbb{S}_R/\mathbb{S}_A) \otimes^\blacksquare_{\mathbb{S}_A,\phi_{tC_p}} \mathbb{S}_A \otimes^\blacksquare \mathrm{THH}_\blacksquare(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) & \longrightarrow & \mathrm{THH}_\blacksquare(\mathbb{S}_R/\mathbb{S}_A)^{t\mathbb{Z}/p} \otimes^\blacksquare \mathrm{ku}_p^{t\mathbb{Z}/p} \\ \simeq\downarrow & & \downarrow \\ \mathrm{THH}_\blacksquare\big((R \otimes^\blacksquare_{A,\phi} A)[\zeta_p]/\mathbb{S}_A[\![q^{1/p}-1]\!]\big) & \dashrightarrow & \mathrm{THH}^\blacksquare(\mathrm{ku}_R/\mathrm{ku}_A)^{t\mathbb{Z}/p} \end{array}$$

and applying $(-)^{tS^1}$ to the dashed arrow. We claim that there is an equivalence

$$\mathrm{TP}_\blacksquare(R \otimes_{\mathbb{Z}_p} \mathbb{F}_p/\mathbb{S}_A) \simeq \mathrm{HP}_\blacksquare(R/A)$$

such that $\overline{\psi}_R$ can be obtained by composing the equivalence with relative Frobenius

$$\phi^{hS^1} : \mathrm{TC}^-_\blacksquare\big((R \otimes^\blacksquare_{\mathbb{Z}_p} \mathbb{F}_p \otimes^\blacksquare_{A,\phi} A)/\mathbb{S}_A\big)[u^{-1}]^\wedge_p \to \mathrm{TP}^\blacksquare(R \otimes^\blacksquare_{\mathbb{Z}_p} \mathbb{F}_p/\mathbb{S}_A)$$

That means, we have the following factor through

$$\begin{array}{ccccc} \mathrm{TC}^-_\blacksquare\big((R \otimes^\blacksquare_{\mathbb{Z}_p} \mathbb{F}_p \otimes^\blacksquare_{A,\phi} A)/\mathbb{S}_A\big)[u^{-1}]^\wedge_p & & \xrightarrow{\quad\overline{\psi}_R\quad} & & \mathrm{HP}_\blacksquare(R/A) \\ & \searrow^{\phi^{hS^1}} & & \nearrow_{\simeq} & \\ & & \mathrm{TP}_\blacksquare(R \otimes_{\mathbb{Z}_p} \mathbb{F}_p/\mathbb{S}_A) & & \end{array}$$

In order to construct the claimed equivalence, we apply Theorem 3.3 and tensor it with $\mathrm{THH}(R/\mathbb{S}_A)$ to form a diagram as follows.

$$\begin{array}{ccc} \mathrm{THH}_\blacksquare(R/\mathbb{S}_A) \otimes^\blacksquare_{j_p} \mathbb{Z}_p & \longrightarrow & \mathrm{THH}_\blacksquare(R/\mathbb{S}_A) \otimes^\blacksquare_{\mathrm{THH}_\blacksquare(\mathbb{Z}_p)} \mathbb{Z}_p \\ \downarrow & & \vdots\downarrow \\ \mathrm{THH}_\blacksquare(R/\mathbb{S}_A) \otimes^\blacksquare_{j_p} \mathrm{THH}(\mathbb{F}_p) & \longrightarrow & \mathrm{THH}_\blacksquare(R/\mathbb{S}_A) \otimes^\blacksquare_{\mathrm{THH}_\blacksquare(\mathbb{Z}_p)} \mathrm{THH}(\mathbb{F}_p) \end{array}$$

All the arrows in the above diagram become equivalence after applying $(-)^{tS^1}$. Therefore, we obtain the right vertical arrow, which is exactly the equivalence $\mathrm{TP}_{\blacksquare}(R \otimes_{\mathbb{Z}_p} \mathbb{F}_p/\mathbb{S}_A) \simeq \mathrm{HP}_{\blacksquare}(R/A)$ we want.

It is then obvious that the factor through goes, since we have

$$\mathrm{THH}_{\blacksquare}(R/\mathbb{S}_A) \simeq \mathrm{THH}_{\blacksquare}(\mathbb{S}_R/\mathbb{S}_A) \otimes^{\blacksquare} \mathrm{THH}(\mathbb{Z}_p)$$

and both ϕ^{hS^1} and $\overline{\psi}_R$ are formed by relative Frobenius.

□

Lemma 3.4. *The modulo $(q-1) = \beta t$ of the map ψ_R^0 agrees with the classical Hodge completion map*

$$\mathrm{dR}_{R/A} \longrightarrow \widehat{\mathrm{dR}}_{R/A} \simeq \mathrm{gr}^0_{\mathrm{ev}}\mathrm{HP}_{\blacksquare}(R/A)$$

Proof strategy. Consider the mod $q-1$ reduction of ψ_R^*, which we denote by $\overline{\psi}_R^*$. Therefore $\overline{\psi}_R^*$ induces a filtered map of Bhatt-Marrow-Scholze even filtration

$$\overline{\psi}_R^* : \mathrm{fil}^*_{\mathrm{BMS}}\mathrm{TC}^-_{\blacksquare}\big((R \otimes^{\blacksquare}_{\mathbb{Z}_p} \mathbb{F}_p \otimes^{\blacksquare}_{A,\phi} A)/\mathbb{S}_A\big)[u^{-1}]^{\wedge}_p \longrightarrow \mathrm{fil}^*_{\mathrm{BMS}}\mathrm{HP}_{\blacksquare}(S/A)$$

By crystalline comparison from [[BS22], Theorem 5.2], we have the zeroth graded piece of the form

$$\overline{\psi}_R^* : \mathrm{dR}_{R/A} \longrightarrow \widehat{\mathrm{dR}}_{R/A} \simeq \mathrm{gr}^0_{\mathrm{ev}}\mathrm{HP}_{\blacksquare}(R/A)$$

We need to show that this map is the desired Hodge completion map. By quasi-syntomic descent, we reduce to the case when both sides are even, thus the filtrations become the double speed Whitehead filtrations. Therefore it suffices to check on the level of static rings. Recall from our assumption of A from Section 1.1, A has a perfectly cover A_∞, which by base change allows us to reduce to the case when A is perfect. Then $\mathrm{A}_{\mathrm{inf}} := \mathrm{W}(S^\flat) \to S$ is surjective.

We then reduce the map to a form of $\mathrm{A}_{\mathrm{crys}} \to \hat{\mathrm{A}}_{\mathrm{crys}}$, which could be checked coincide with the completion map by some unique characteristics.

□

Corollary 3.5. *After rationalization, we have a commutative diagram*

$$\begin{array}{ccc} q-\mathrm{dR}_{R/A}[p^{-1}]^{\wedge}_{q-1} & \longrightarrow & \mathrm{gr}^0_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)[p^{-1}]^{\wedge}_{q-1} \\ \downarrow & & \downarrow\simeq \\ \mathrm{dR}_{R/A}[p^{-1}][\![q-1]\!] & \longrightarrow & \mathrm{dR}_{R/A}[p^{-1}]^{\wedge}_{\mathrm{Hdg}}[\![q-1]\!] \end{array}$$

where the bottom map consists with the natural Hodge completion map.

Proof. Note that the square is commutative as $\mathbb{Z}_p^\times$-equiavriant modules. Therefore, we could apply a general fact that

$$M \simeq M[\![q-1]\!]^{h\mathbb{Z}_p^\times} \otimes_{\mathbb{Z}_p^\times} \mathbb{Z}_p$$

to kill $[\![q-1]\!]$ summand, then compare with the modulo $q-1$ case with the above lemma. □

Proof of Theorem 3.1. The equivalence

$$\mathrm{fil}^*_{\mathrm{qHdg}} q-\widehat{\mathrm{dR}}_{R/A} \xrightarrow{\simeq} \Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

can be checked modulo $q-1$. For the statement with regard to $\mathrm{fil}^*_{\mathrm{qHdg}} q-\mathrm{dR}_{R/A}\otimes_{\mathbb{Z}_p[\beta][\![t]\!]}\mathbb{Z}_p[\beta]$, note that $\mathrm{fil}^*_{\mathrm{qHdg}} q-\mathrm{dR}_{R/A}\otimes_{\mathbb{Z}_p[\beta][\![t]\!]}\mathbb{Z}_p[\beta]$ fits into a pullback diagram

$$\begin{array}{ccc} \mathrm{fil}^*_{\mathrm{qHdg}} q-\mathrm{dR}_{R/A}\otimes_{\mathbb{Z}_p[\beta][\![t]\!]}\mathbb{Z}_p[\beta] & \longrightarrow & \Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{HC}^-_{\blacksquare}(R/A) \\ \downarrow & & \downarrow \\ \mathrm{dR}_{R/A} & \longrightarrow & \mathrm{gr}^0_{\mathrm{ev}}\mathrm{HC}^-_{\blacksquare}(R/A) \end{array}$$

where the bottom by Lemma 3.4 is the completion map. By the fact that any filtration is the pullback of its completion, we finish the proof of (2). (3) follows analogously by Corollary 3.5. □

Remark 3.6 (Why $p=2$ fails)**.** There are different places the above discussion fails when $p=2$. In this case Theorem 4.1 is still in conjecture, and Theorem 3.3 is provabaly wrong. As we can see in the next section, the difficulty will happen in a few places:

(1) In the process of proving (1) in Section 4.1, we will need to consider the mod p spectrum $j_{p,0}/p$. When $p=2$, the resulting spectrum will not admit any associative structure, and the discussion fails.
(2) In Lemma 4.5, one requires $p\geqslant 3$ to guarantee that the 1-truncation concentrates in $\leqslant 2p-3$.

Remark 3.7 (Amendments for $p=2$)**.** There is an equivalence

$$\tau_{\geqslant 0}(\mathrm{ku}_p^{t\mathbb{Z}/p}) \simeq \mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!])$$

of $\mathbb{E}_1$-rings, which is an unpublished result of Nikolaus. Using this, one can construct the comparison map for $p=2$. If we adopt Assumption 1.2, then we can circumvent Theorem 3.3. In this case Theorem 3.1 is still true.

3.2. **The global case.** We now attempt to piece everything together. We are going to form the analogy of the results in the previous section to the global case. Therefore we will need new assumptions of our rings.

Assumption 3.8. In this section, we require A and R to satisfy the following conditions.

(1) The ring A is perfectly-covered and for all primes p, $\hat{A}_p$ should satisfy Assumption 1.1. We use $\mathbb{S}_{\hat{A}_p}$ to denote their p-complete spherical lifts.
(2) R is quasi-lci over A and has bounded p-torsion. For every prime p, its p-completion $\hat{R}_p$ satisfies either Assumption 1.2 or Assumption 1.3. We use $\mathbb{S}_{\hat{R}_p}$ to denote the p-complete spherical lifts.
(3) Additionally, in order to fix the flaws that happen when $p=2$, we require $\hat{R}_2$ to satisfy Assumption 1.2.

By gluing through $A\otimes\mathbb{Q}$ and $R\otimes\mathbb{Q}$ together, one get a connective $\mathbb{E}_\infty$-ring $\mathbb{S}_A$ and a connective $\mathbb{E}_1$-ring $\mathbb{S}_R$ such that $\mathbb{S}_A\otimes\mathbb{Z}\simeq A$ and $\mathbb{S}_R\otimes\mathbb{Z}\simeq R$.

We first need to define the solid even filtration in the profinite case from the p-complete case. Let $\hat{A}$ and $\hat{R}$ denote the profinite-completion of A and R. Let k be a connective even

$\mathbb{E}_\infty$-ring. We define $k_{\hat{A}} := k \otimes^{\blacksquare} \prod_p \mathbb{S}_{\hat{A}_p}$ and $k_{\hat{R}} := k \otimes^{\blacksquare} \prod_p \mathbb{S}_{\hat{R}_p}$. We want to define even filtration

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}})$$

First, we construct this even filtration for two special cases.

Construction 3.9. We consider two cases first.

(1) Suppose all the primes satisfy Assumption 1.2, we denote $\mathbb{S}_{\hat{R}_p} \to \mathbb{S}_{\hat{R}^\bullet_{p,\infty}}$ to be each cosimplicial resolution and $k_{\hat{R}^\bullet_\infty} := k \otimes^{\blacksquare} \prod_p \mathbb{S}_{\hat{R}_p}$. We define the filtration by

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}}) := \lim_{\Delta} \tau_{\geqslant 2*} \mathrm{THH}_{\blacksquare}(k_{\hat{R}^\bullet_\infty}/k_{\hat{A}})$$

(2) Suppose all the primes satisfy Assumption 1.3. In this case $k_{\hat{R}}/k_{\hat{A}}$ become an $\mathbb{E}_1$ algebra. In this case we directly define $\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}})$ to be the solid even filtration over itself.

(3) In general, let P_1 and P_2 denote the set of primes that satisfy Assumption 1.2 and Assumption 1.3 respectively. Denote $k_{\hat{R},1} := \prod_{p \in P_1} k_{\hat{R}_p}$ and $k_{\hat{R},2} := \prod_{p \in P_2} k_{\hat{R}_p}$. Then we define

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}}) = \mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R},1}/k_{\hat{A}}) \times \mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R},2}/k_{\hat{A}})$$

Remark 3.10. We could move every result to the profinite case verbatismly. Especially, in the profinite case we still have a cosimplicial resolution

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}}) \simeq \lim_{\Delta} \tau_{\geqslant 2*} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}} \otimes^{\blacksquare} \mathbb{S}_{\hat{P}^\bullet})$$

which is analogous to that of

We can also define the profinite even filtration of $\mathrm{TC}^-_{\blacksquare}$ and $\mathrm{TP}_{\blacksquare}$.

Definition 3.11. We define the profinite even filtration of $\mathrm{TC}^-_{\blacksquare}$ and $\mathrm{TP}_{\blacksquare}$ to be

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{TC}^-_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}}) := \left(\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}})\right)^{h\mathbb{T}_{\mathrm{ev}}}$$

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{TP}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}}) := \left(\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(k_{\hat{R}}/k_{\hat{A}})\right)^{t\mathbb{T}_{\mathrm{ev}}}$$

We first explain a lemma, which shows that the profinite even filtration defined above is indeed the "profinite complete" of the even filtrations in our case when $k = \mathrm{ku}$.

Lemma 3.12. *We have canonical equivalences*

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(\mathrm{ku}_{\hat{R}}/\mathrm{ku}_{\hat{A}}) \simeq \prod_p \mathrm{fil}^*_{\mathrm{ev}} \mathrm{THH}_{\blacksquare}(\mathrm{ku}_{\hat{R}_p}/\mathrm{ku}_{\hat{A}_p})$$

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_{\hat{R}}/\mathrm{ku}_{\hat{A}}) \simeq \prod_p \mathrm{fil}^*_{\mathrm{ev}} \mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_{\hat{R}_p}/\mathrm{ku}_{\hat{A}_p})$$

$$\mathrm{fil}^*_{\mathrm{ev}} \mathrm{TP}_{\blacksquare}(\mathrm{ku}_{\hat{R}}/\mathrm{ku}_{\hat{A}}) \simeq \prod_p \mathrm{fil}^*_{\mathrm{ev}} \mathrm{TP}_{\blacksquare}(\mathrm{ku}_{\hat{R}_p}/\mathrm{ku}_{\hat{A}_p})$$

Proof. We let $\mathrm{ku}_A := \mathrm{ku} \otimes \mathbb{S}_A$ and $\mathrm{ku}_R := \mathrm{ku} \otimes \mathbb{S}_R$. By properties of solid tensor product, one can show that ku. Since solid THH takes module to the completion (both of the p-completion and profinite completion) of its discrete THH, one has

$$\mathrm{THH}_{\blacksquare}(\mathrm{ku}_{\hat{R}}/\mathrm{ku}_{\hat{A}}) \simeq \mathrm{THH}(\mathrm{ku}_R/\mathrm{ku}_A)^{\wedge} \simeq \prod_p \mathrm{THH}(\mathrm{ku}_R/\mathrm{ku}_A)^{\wedge}_p \simeq \prod_p \mathrm{THH}_{\blacksquare}(\mathrm{ku}_{\hat{R}_p}/\mathrm{ku}_{\hat{A}_p})$$

□

Now we apply the filtration operation. Since in both Assumption 1.2 and Assumption 1.3 the even filtration could be written as a cosimplicial limit, which commutes with product, we finish the proof of the first equavalence.

The second equivalence follows immediately, since the homotopy fixed points operation commutes with all limits. To see the third equivalence, one need to show that the synthetic homotopy orbit functor $(-)_{h\mathbb{T}_{\mathrm{ev}}} = \mathbb{S}_{\mathrm{ev}} \otimes^{\blacksquare}_{\mathbb{T}_{\mathrm{ev}}} (-)$ commutes with the infinite product $\prod_p$. This could be done by checking after modulo β, which reduces to check for $\mathrm{fil}^*_{\mathrm{HKR}}\mathrm{HH}_{\blacksquare}(\hat{R}/\hat{A})$. It thus suffices to check on graded pieces $\prod_p \mathrm{gr}^{\mathrm{n}}_{\mathrm{ev}}\mathrm{HH}_{\blacksquare}(\hat{R}_p/\hat{A}_p) \simeq \prod_p \mathrm{gr}^{\mathrm{n}}_{\mathrm{ev}}\Omega^*(\hat{R}_p/\hat{A}_p)$. Since we require that R is quasi-lci over A, each graded piece concentrates in finite range. Therefore, only finite many cells of BS^1 will contribute to the homotopy orbit $(-)_{h\mathbb{T}_{\mathrm{ev}}}$. Thus it commutes with the infinte product $\prod_p$.

We are going to define the global filtrations, which are obtained by pull back the profinte even filtration along the profinite completion map.

Definition 3.13. We define the even filtration $\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}(\mathrm{ku}_R/\mathrm{ku}_A)$ to be the pullback

$$\begin{array}{ccc} \mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}(\mathrm{ku}_R/\mathrm{ku}_A) & \longrightarrow & \mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_{\blacksquare}(\mathrm{ku}_{\hat{R}}/\mathrm{ku}_{\hat{A}}) \\ \downarrow & & \downarrow \\ \mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}(\mathrm{ku}_R \otimes \mathbb{Q}/\mathrm{ku}_A \otimes \mathbb{Q}) & \longrightarrow & \mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_{\blacksquare}(\mathrm{ku}_{\hat{R}} \otimes^{\blacksquare} \mathbb{Q}/\mathrm{ku}_{\hat{A}} \otimes^{\blacksquare} \mathbb{Q}) \end{array}$$

The bottom map is the profinite completion map

$$\mathrm{fil}^*_{\mathrm{ev}}\mathrm{HH}(R/A) \otimes \mathbb{Q}[\beta]_{\mathrm{ev}} \to \mathrm{fil}^*_{\mathrm{ev}}\mathrm{HH}_{\blacksquare}(\hat{R}/\hat{A}) \otimes \mathbb{Q}[\beta]_{\mathrm{ev}}$$

Here we again use the fact that solid THH takes modules to the completion of there discrete THH.

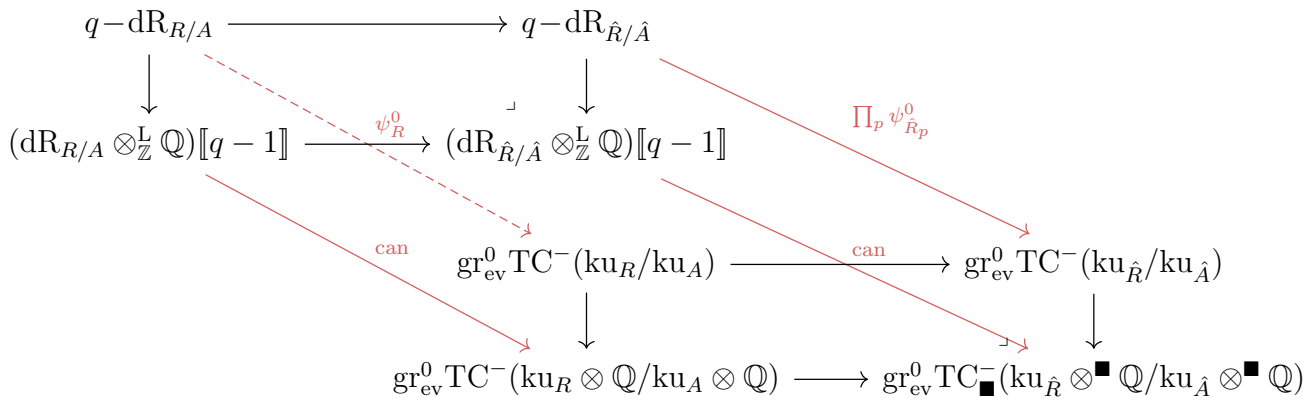
Lemma 3.14. *If R satisfies Assumption 1.3 for all primes p, then the pullback filtration $\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}(\mathrm{ku}_R/\mathrm{ku}_A)$ agrees with the solid even filtration $\mathrm{fil}^*_{\mathrm{ev}}\mathrm{THH}_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$.*

Proof. By checking that both of the filtrations can be computed by cosimplicial resolution. □

We now construct the global comparison map. Firstly, by the definition of the global q-de Rham complex, one has a pullback square

$$\begin{array}{ccc} q\text{-}\mathrm{dR}_{R/A} & \longrightarrow & q\text{-}\mathrm{dR}_{\hat{R}/\hat{A}} \\ \downarrow & & \downarrow \\ (\mathrm{dR}_{R/A} \otimes^{\mathrm{L}}_{\mathbb{Z}} \mathbb{Q})[\![q-1]\!] & \longrightarrow & (\mathrm{dR}_{\hat{R}/\hat{A}} \otimes^{\mathrm{L}}_{\mathbb{Z}} \mathbb{Q})[\![q-1]\!] \end{array}$$

We want to map this pullback square to another pullback square



Here the map can denotes the canonical Hodge completion map. The bottem square is already commutative. We need to show the right square is commutative. This could be proven with the same fashion as Corollary 3.5.

Definition 3.15. We define the **global q-Hodge filtration** of $q-\mathrm{dR}_{R/A}$ to be the pullback of graded $\mathbb{Z}[\beta][\![t]\!]$-modules

$$\begin{array}{ccc} \mathrm{fil}^*_{\mathrm{qHdg}}q-\mathrm{dR}_{R/A} & \longrightarrow & \Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A) \\ \downarrow & & \downarrow \\ q-\mathrm{dR}_{R/A} & \longrightarrow & \mathrm{gr}^0_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A) \end{array}$$

Analogously to Construction 2.12, here we identify $\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}(\mathrm{ku}^{hS^1})\simeq\pi_{2*}(\mathrm{ku}^{hS^1})\simeq\mathbb{Z}[\beta][\![t]\!]$ with $(q-1)^*\mathbb{Z}[\![q-1]\!]$, with t the filtration degree and β identified with $q-1$ in degree 1.

We have a similar result in global case.

Theorem 3.16. *Let A and R be as in Assumption 3.8. Then we have*

(1) The map above induces an equivalence

$$\mathrm{fil}^*_{\mathrm{qHdg}}q-\widehat{\mathrm{dR}}_{R/A}\xrightarrow{\simeq}\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

(2) After modulo $\beta=q-1$ in degree 1, we got the equivalence

$$\mathrm{fil}^*_{\mathrm{qHdg}}q-\mathrm{dR}_{R/A}\otimes_{\mathbb{Z}[\beta][\![t]\!]}\mathbb{Z}[\beta]\xrightarrow{\simeq}\mathrm{fil}^*_{\mathrm{Hdg}}\mathrm{dR}_{R/A}$$

(3) We also have an equivalence

$$\mathrm{fil}^*_{\mathrm{qHdg}}q-\mathrm{dR}_{R/A}\otimes\mathbb{Q}\xrightarrow{\simeq}\mathrm{fil}^*_{\mathrm{Hdg},q-1}(\mathrm{dR}_{R/A}\otimes\mathbb{Q})[\![q-1]\!]$$

Proof. This result follows verbatim from Theorem 3.1. □

4. Proof of the main ingredient

In this section, we prove our main ingredient.

Theorem 4.1. *There is a commutative diagram of $\mathbb{E}_\infty$ rings for $p > 2$.*

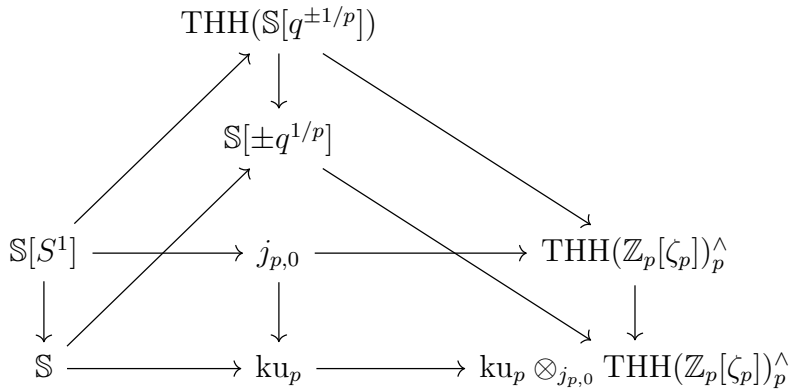
$$\begin{array}{ccccc}
j_{p,0} & \longrightarrow & \tau_{\geqslant 0} j_{p,0}^{t\mathbb{Z}/p} & \xrightarrow{\simeq} & \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \\
\downarrow & \ulcorner & \downarrow & & \downarrow \\
\mathrm{ku}_p & \longrightarrow & \tau_{\geqslant 0}\mathrm{ku}_p^{t\mathbb{Z}/p} & \xrightarrow{\simeq} & \mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!]) \\
\downarrow & & \downarrow & & \downarrow \\
\mathbb{Z}_p & \longrightarrow & \tau_{\geqslant 0}\mathbb{Z}_p^{t\mathbb{Z}/p} & \xrightarrow{\simeq} & \mathrm{THH}(\mathbb{F}_p)
\end{array}$$

where $j_{p,0}$ is $\tau_{\geqslant 0}(\mathrm{KU}_p^{h\mathbb{Z}})$ with $\mathbb{Z} \subseteq \mathbb{Z}_p$ generated by $1+p$ and $\mathbb{S}[\![q^{1/p}-1]\!]$ is $\mathbb{S}[\![q^{1/p}-1]\!] \simeq \lim \mathbb{S}[q^{1/p}]/(q^\alpha - 1) \simeq \lim \mathbb{S}[\mathbb{Z}/p^\alpha]$.

4.1. **The pushout $\mathrm{ku}_p \otimes_{j_{p,0}} \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \simeq \mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!])$.** We will begin with the equivalence

$$\mathrm{ku}_p \otimes_{j_{p,0}} \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \simeq \mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!])$$

Our strategy to prove this is to fit everything in a big pushout diagram. We use L to denote the pushout $\mathrm{ku}_p \otimes_{j_{p,0}} \mathrm{THH}(\mathbb{Z}_p[\zeta_p])$, and we show that every square in the following diagram is a pushout square.



Then we will have

$$\begin{aligned}
\mathrm{ku}_p \otimes_{j_{p,0}} \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \simeq & \mathbb{S} \otimes_{\mathbb{S}[S^1]} \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \\
\simeq & \mathbb{S}[q^{\pm 1/p}] \otimes_{\mathrm{THH}(\mathbb{S}[q^{\pm 1/p}])} \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \\
\simeq & \mathbb{S}[\![q^{1/p}-1]\!] \otimes_{\mathrm{THH}(\mathbb{S}[\![q^{1/p}-1]\!])} \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \\
\simeq & \mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}[\![q^{1/p}-1]\!])
\end{aligned}$$

Therefore, we need to check that

(1) We have a pushout $\mathbb{S} \otimes_{\mathbb{S}[S^1]} j_{p,0} \simeq \mathrm{ku}_p$.
(2) We have a factor through $\mathbb{S}[S^1] \to \mathrm{THH}(\mathbb{S}[q^{\pm 1/p}]) \to \mathrm{THH}(\mathbb{Z}_p[\zeta_p])$.
(3) We have a pushout $\mathbb{S} \otimes_{\mathbb{S}[S^1]} \mathrm{THH}(\mathbb{S}[q^{\pm 1/p}]) \simeq \mathbb{S}[q^{\pm 1/p}]$.

We will also need to check that there is a pushout diagram

$$\begin{array}{ccc} j_{p,0} & \longrightarrow & \tau_{\geqslant 0} j_{p,0}^{t\mathbb{Z}/p} \\ \downarrow & \ulcorner & \downarrow \\ \mathrm{ku}_p & \longrightarrow & \tau_{\geqslant 0}\mathrm{ku}_p^{t\mathbb{Z}/p} \end{array}$$

Moreover, we have to explain why we have an equivalence.

$$\tau_{\geqslant 0} j_{p,0}^{t\mathbb{Z}/p} \longrightarrow \mathrm{THH}(\mathbb{Z}_p[\zeta_p])$$

We first prove (3), which is almost obvious.

Proof of (3). We first note that for any infinite loopspace G, one has a well-known equivalence

$$\mathrm{THH}(\mathbb{S}[G]) \simeq \Sigma^\infty_+ LBG$$

Especially, if we let G be S^1, then $* \to S^1$ induces a map $\mathbb{S}[S^1] \to \mathrm{THH}(\mathbb{S}[\Omega S^1]) \simeq \mathrm{THH}(\mathbb{S}[q^{\pm 1/p}])$. On space level we have a homotopy fiber

$$S^1 \longrightarrow LS^1 \longrightarrow \Omega S^1$$

where the second map is projection on each connected components. The fiber on space level induces a pushout square

$$\begin{array}{ccc} \mathbb{S}[S^1] & \longrightarrow & \mathrm{THH}(\mathbb{S}[q^{\pm 1/p}]) \\ \downarrow & \ulcorner & \downarrow \\ \mathbb{S} & \longrightarrow & \mathbb{S}[q^{\pm 1/p}] \end{array}$$

which finish the construction. □

To prove the part (1), we need to use ∞-group of units from Section 1.3. We begin with an observation.

Lemma 4.2. *There is a commutative square of spectra*

$$\begin{array}{ccc} \Sigma\mathbb{F}_p & \longrightarrow & \mathfrak{gl}_1(j_{p,0}) \\ \downarrow & & \downarrow \\ \Sigma^2\mathbb{Z}_p & \longrightarrow & \mathfrak{gl}_1(\mathrm{ku}_p) \end{array}$$

where the left vertical map is the Bockstein, the bottom map is the detective map of $\beta \in \pi_2(\mathrm{ku}_p)$.

Proof. The bottom map is $\mathbb{Z}_p^\times$-equivariant, so we can apply the homotopy fixed points functor $(-)^{h\mathbb{Z}}$ to

$$\Sigma^2\mathbb{Z}_p \longrightarrow \mathfrak{gl}_1(\mathrm{ku}_p)$$

The homptopy fixed points $\mathbb{Z}_p^{h\mathbb{Z}}$ is $\Sigma^{-1}\mathbb{F}_p$ with canonical map $\Sigma^2\mathbb{Z}_p^{h\mathbb{Z}} \to \Sigma^2\mathbb{Z}_p$ identified with the Bockstein. And the homotopy fixed points $\mathfrak{gl}_1(\mathrm{ku}_p)^{h\mathbb{Z}}$ is identified with $\mathfrak{gl}_1(j_{p,0})$, since $\mathfrak{gl}_1(-)$ is a left adjoint which commutes with homotopy fixed points. □

We can precompose the diagram with $\Sigma\mathbb{Z} \to \Sigma\mathbb{F}_p$, then the map $\Sigma\mathbb{Z} \to \mathfrak{gl}_1(\mathrm{ku}_p)$ will be null-homotopic.

$$\begin{array}{ccccc} \Sigma\mathbb{Z} & \longrightarrow & \Sigma\mathbb{F}_p & \longrightarrow & \mathfrak{gl}_1(j_{p,0}) \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \Sigma^2\mathbb{Z}_p & \longrightarrow & \mathfrak{gl}_1(\mathrm{ku}_p) \end{array}$$

The above map detects a generator in $\pi_1(j_{p,0})$. We can apply $\mathbb{S}[-]$ to the outer square. We obtain a commutative square of $\mathbb{E}_\infty$-rings, which is exactly what we want.

$$\begin{array}{ccc} \mathbb{S}[S^1] & \longrightarrow & j_{p,0} \\ \downarrow & & \downarrow \\ \mathbb{S} & \longrightarrow & \mathrm{ku}_p \end{array}$$

Now we prove (1), which states that the above diagram is a pushout diagram of $\mathbb{E}_\infty$-rings.

Proof of (1). We need to show that $\pi_*(j_{p,0} \otimes_{\mathbb{S}[S^1]} \mathbb{S}) \cong \pi_*(\mathrm{ku}_p)$. Therefore, we need to calculate the former homotopy group. Since everything is p-complete, we reduce to the mod p case. $\mathbb{S}/p$ admits an $\mathbb{A}_2$-structure, then after mod p every homotopy group admits a structure of associative ring. We first compute $\pi_*(j_{p,0})$.

We consider ψ^{1+p} and id to be two endomorphisms of ku_p, where ψ^{1+p} is obtained through action by $1+p$. The endomorphism $\psi^{1+p} - \mathrm{id}$ essentially factors through $\tau_{\geqslant 2}\mathrm{ku}_p$. We define this new map $\mathrm{ku}_p \to \tau_{\geqslant 2}\mathrm{ku}_p$ to be $\psi_\circ^{1+p}$. By definition of homotopy fixed points, one has a fiber sequence

$$j_{p,0} \longrightarrow \mathrm{ku}_p \xrightarrow{\psi_\circ^{1+p}} \tau_{\geqslant 2}(\mathrm{ku}_p)$$

The map $\psi_\circ^{1+p}$ takes the generator $u \in \pi_2(\mathrm{ku}_p)$ to pu. By analizing along the fiber sequence after mod p, we obtain that

$$\pi_*(j_{p,0}/p) \cong \mathbb{F}_p[u] \otimes_{\mathbb{F}_p} \Lambda(\delta)$$

where u is a generator in degree 2 which maps to the Bott element in ku_p, δ is a generator in degree 1, which is the image of S^1.

Therefore, we obtain $\pi_*(j_{p,0}/p \otimes_{\mathbb{S}[S^1]} \mathbb{S}) \cong \mathbb{F}_p(u)$, which coincides with the mod p homotopy of ku_p, as desired.

□

Before proving the remaining part (2), we can immediately prove the pushout square in Theorem 4.1.

Lemma 4.3. *There is a pushout diagram of $\mathbb{E}_\infty$-rings*

$$\begin{array}{ccc} j_{p,0} & \longrightarrow & \tau_{\geqslant 0}(j_{p,0}^{t\mathbb{Z}/p}) \\ \downarrow & & \downarrow \\ \mathrm{ku}_p & \longrightarrow & \tau_{\geqslant 0}(\mathrm{ku}_p^{t\mathbb{Z}/p}) \end{array}$$

Proof. Due to the previous proof, it suffices to check that we have a pushout diagram

$$\begin{array}{ccc} \mathbb{S}[S^1] & \longrightarrow & \tau_{\geqslant 0}(j_{p,0})^{t\mathbb{Z}/p} \\ \downarrow & & \downarrow \\ \mathbb{S} & \longrightarrow & \tau_{\geqslant 0}(\mathrm{ku}_p)^{t\mathbb{Z}/p} \end{array}$$

We use a strategy similar to that in the proof of (1). We apply $(-)^{t\mathbb{Z}/p}$ to the fiber sequence $j_{p,0} \to \mathrm{ku}_p \to \tau_{\geqslant 0}\mathrm{ku}_p$. Then we obtain a new fiber sequence

$$\tau_{\geqslant 0}(j_{p,0})^{t\mathbb{Z}/p} \longrightarrow \tau_{\geqslant 0}(\mathrm{ku}_p)^{t\mathbb{Z}/p} \xrightarrow{\psi_{\circ\circ}^{1+p}} \tau_{\geqslant 2}(\tau_{\geqslant 2}(\mathrm{ku}_p)^{t\mathbb{Z}/p})$$

The induced map $\psi_{\circ\circ}^{1+p}$ sends $u \in \tau_{\geqslant 0}(\mathrm{ku}_p)^{t\mathbb{Z}/p}$ to $\frac{p}{\zeta_p - 1}u$. To see this, we consider the map of fiber sequences.

$$\begin{array}{ccc} j_{p,0} & \longrightarrow & \tau_{\geqslant 0}(j_{p,0})^{t\mathbb{Z}/p} \\ \downarrow & & \downarrow \\ \mathrm{ku}_p & \longrightarrow & \tau_{\geqslant 0}(\mathrm{ku}_p)^{t\mathbb{Z}/p} \\ \downarrow & & \downarrow{\scriptstyle \psi_{\circ\circ}^{1+p}} \\ \tau_{\geqslant 2}(\mathrm{ku}_p) & \longrightarrow & \tau_{\geqslant 2}(\tau_{\geqslant 2}(\mathrm{ku}_p)^{t\mathbb{Z}/p}) \end{array}$$

The middle horizontal map sends the generator $u \in \pi_2(\mathrm{ku}_p)$ to $(\zeta_p - 1)u$ and the bottom horizontal map sends u to u. Tracing counterclockwise, we obtain the action of $\psi_{\circ\circ}^{1+p}$ on homotopy groups.

We can now analyze the fiber sequence. By base change to $\mathbb{F}_p$, one computes that

$$\pi_*(\tau_{\geqslant 0}j_{p,0}^{t\mathbb{Z}/p} \otimes_{\mathbb{Z}_p[\zeta_p]} \mathbb{F}_p) \cong \mathbb{F}_p[u] \otimes \Lambda(\epsilon)$$

To identify the homotopy group of $\pi_*(\tau_{\geqslant 0}(j_{p,0}^{t\mathbb{Z}/p}) \otimes_{\mathbb{S}[S^1]} \mathbb{S})$ with $\pi_*(\tau_{\geqslant 0}(\mathrm{ku}_p^{t\mathbb{Z}/p}))$, we need to identify δ with ϵ. This is because he bottom horizontal map $\pi_2(\tau_{\geqslant 2}(\mathrm{ku}_p)) \to \pi_2(\tau_{\geqslant 2}(\mathrm{ku}_p^{t\mathbb{Z}/p}))$ sends u to u and both $\pi_2(\tau_{\geqslant 2}(\mathrm{ku}_p)) \to \pi_1(j_{p,0})$ and $\pi_2(\tau_{\geqslant 2}(\mathrm{ku}_p^{t\mathbb{Z}/p})) \to \pi_1(\tau_{\geqslant 0}(j_{p,0}^{t\mathbb{Z}/p}))$ are surjective.

□

We now prove (2). That means, we need to show that we need to construct a factor-through diagram of cyclotomic $\mathbb{E}_\infty$-rings.

$$\begin{array}{ccc} \mathbb{S}[S^1] & \xrightarrow{\delta} & j_{p,0} \\ \downarrow & & \downarrow \\ \mathrm{THH}(\mathbb{S}[q^{\pm 1}]) & \longrightarrow & \mathrm{THH}(\mathbb{Z}_p[\zeta_p]) \end{array}$$

Since everything is in the cyclotomic world and $\mathbb{S}[S^1]$ and $j_{p,0}$ admit trivial cyclotomic structures, it suffices to check on TC-level.

Lemma 4.4. *There is a commutative square of $\mathbb{E}_\infty$-rings*

$$\begin{array}{ccc} \mathbb{S}[S^1] & \xrightarrow{\delta} & j_{p,0} \\ \downarrow & & \downarrow \\ \mathrm{TC}(\mathbb{S}[q^{\pm 1}]) & \longrightarrow & \mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \end{array}$$

The map $j_{p,0} \to \mathrm{TC}(\mathbb{Z}_p[\zeta_p])$ is constructed as follows. Since KU_p is a model for $K(1)$-local sphere, there is a standard map $\mathrm{KU}_p \to \mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p))$. This standard map is $\mathbb{Z}_p^\times$-equivariant. Therefore, after applying $\tau_{\geqslant 0}((-)^{h\mathbb{Z}})$, we obtain an $\mathbb{F}_p^\times$-equivariant map

$$j_{p,0} \longrightarrow \mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p]))$$

By Lichtenbaum-Quillen conjecture, there is an equivalence

$$\tau_{\geqslant 2}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \simeq \tau_{\geqslant 2}\mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p]))$$

It suffices to construct a factor through

$$\begin{array}{ccccc} j_{p,0} & \longrightarrow & \tau_{\geqslant 0}\mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p])) & \longrightarrow & \tau_{[0,1]}\mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p])) \\ & \searrow & & \nearrow & \\ & & \tau_{[0,1]}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) & & \end{array}$$

We have an equivalence $\tau_{[0,1]}j_{p,0} \simeq \tau_{[0,1]}\mathbb{S}[B\mathbb{Z}/p]$ which is induced by the map $j_{p,0} \to \mathbb{S}[B\mathbb{Z}/p]$ from Lemma 4.2. The map $\tau_{[0,1]}\mathbb{S}[B\mathbb{Z}/p] \to \tau_{[0,1]}\mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p]))$ determines and is determined by the inclusion $\mathbb{Z}/p \to \mathrm{GL}(\mathbb{Z}_p[\zeta_p])$. Therefore, the map

$$j_{p,0} \to \tau_{[0,1]}\mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p]))$$

essentially factors through $\mathrm{K}(\mathbb{Z}_p[\zeta_p])$. We construct the factor through as follows.

$$\begin{array}{ccccc} j_{p,0} & \longrightarrow & \tau_{\geqslant 0}\mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p])) & \longrightarrow & \tau_{[0,1]}\mathrm{L}_{K(1)}(\mathrm{TC}(\mathbb{Z}_p[\zeta_p])) \\ \downarrow & & & & \uparrow \\ \tau_{[0,1]}\mathbb{S}[B\mathbb{Z}/p] & \longrightarrow & \mathrm{K}(\mathbb{Z}_p[\zeta_p]) & \longrightarrow & \tau_{[0,1]}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \end{array}$$

We begin with a few lemmas.

Lemma 4.5. *Let R be a p-complete $\mathbb{E}_\infty$-ring which is concentrated in degree $\leqslant 2p-3$. Let $f : \mathbb{S}[S^1] \to R$ be an $\mathbb{E}_\infty$-ring map, which determines an element $\epsilon \in \pi_1(R)$. Then f is fully determined by ϵ.*

Proof. The map $\mathbb{S}[S^1] \to R$ is fully determined by $\Sigma\mathbb{Z} \to \mathfrak{gl}_1(R)$ with $\mathfrak{gl}_1(R)$ concentrate in degree $\leqslant 2p-3$. We deduce that

$$\mathrm{Map}(\Sigma\mathbb{Z}, \mathfrak{gl}_1(R)) \simeq \mathrm{Map}(\tau_{\leqslant 2p-3}(\Sigma\mathbb{Z}), \mathfrak{gl}_1(R)) \simeq \mathrm{Map}(\tau_{\leqslant 2p-3}(\Sigma\mathbb{S}), \mathfrak{gl}_1(R)) \simeq \mathrm{Map}(\Sigma\mathbb{S}, \mathfrak{gl}_1(R))$$

which shows that the map $\Sigma\mathbb{Z} \to \mathfrak{gl}_1(R)$ is fully determined by the map that detects $\epsilon \in \pi_1(R)$. □

We can now prove Lemma 4.4 for 1-truncated case.

Lemma 4.6. *There is a commutative diagram of $\mathbb{E}_\infty$-rings*

$$\begin{array}{ccc} \mathbb{S}[S^1] & \xrightarrow{\delta} & j_{p,0} \\ \downarrow & & \downarrow \\ \mathrm{TC}(\mathbb{S}[q^{\pm 1}]) & \longrightarrow & \tau_{\leqslant 1}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \end{array}$$

Proof. Using the above lemma, this is by checking generators in $\pi_1(\mathrm{TC}(\mathbb{Z}_p[\zeta_p]))$. □

To finish the proof of Lemma 4.4, we will need some comparison consequence of $K(1)$-localization.

Lemma 4.7. *Let R be a p-complete $E(n)$-local $\mathbb{E}_\infty$-ring, and let F be the fiber of $\mathfrak{gl}_1(R) \to \mathrm{L}_{K(1)\vee\cdots\vee K(n)}\mathfrak{gl}_1(R)$. Then we have*

(1) $\mathfrak{gl}_1(R) \to \mathrm{L}_n\mathfrak{gl}_1(R)$ is injective on π_{n+1}.
(2) F is $n+1$-truncated.
(3) $\pi_{n+1}(F)$ is torsion-free.

Proof. The first claim follows from [[AHR10]; Theorem 4.11], which shows that the fiber F' of $\mathfrak{gl}_1(R) \to \mathrm{L}_n\mathfrak{gl}_1(R)$ is n-truncated. In fact, we will not use this claim.

In order to show that $\pi_{n+1}(F)$ is torsion-free, we consider the following commutative diagram.

$$\begin{array}{ccc} & \mathrm{L}_n(\mathfrak{gl}_1(R)) & \\ \nearrow & & \searrow \\ \mathfrak{gl}_1(R) & \longrightarrow & \mathrm{L}_{K(1)\vee\cdots\vee K(n)}\mathfrak{gl}_1(R) \end{array}$$

If we denote the fiber of $\mathrm{L}_n(\mathfrak{gl}_1(R)) \to \mathrm{L}_{K(1)\vee\cdots\vee K(n)}\mathfrak{gl}_1(R)$ by F'', then along the diagram we have the fiber sequence

$$F' \longrightarrow F \longrightarrow F''$$

The homotopy groups of F'' are $\mathbb{Q}$-vector spaces. To see this, we note that there is a pullback diagram of endofunctors of spectra

$$\begin{array}{ccc} \mathrm{L}_n & \longrightarrow & \mathrm{L}_{K(1)\vee\cdots\vee K(n)} \\ \downarrow & \lrcorner & \downarrow \\ \mathrm{L}_{\mathbb{Q}} & \longrightarrow & \mathrm{L}_{\mathbb{Q}}\mathrm{L}_{K(1)\vee\cdots\vee K(n)} \end{array}$$

The homotopy groups of F'' then can be computed along the bottom arrow applied to $\mathfrak{gl}_1(R)$, thus they must be $\mathbb{Q}$-vector spaces. The third claim is now clear since $\pi_{n+1}(F)$ maps to $\pi_{n+1}(F'')$ injectively.

Finally, to finish the proof, we need to show that F is $n+1$-truncated. Since F' is n-truncated, it suffices to show that F'' is $n+1$-truncated.

We note that $\mathrm{L}_n \simeq \mathrm{L}_{K(0)\vee K(1)\vee\cdots\vee K(n)} \to \mathrm{L}_{K(1)\vee\cdots\vee K(n)}$ is given by p-completion. Since in our case everything is p-complete, we have isomorphisms

$$\pi_i(R) \cong \pi_i(\mathfrak{gl}_1(R)) \cong \pi_i(\mathrm{L}_n\mathfrak{gl}_1(R))$$

for $i \geqslant n+2$ and everything is (derived) p-complete. By general universal coefficient theorem, there is a short exact sequence

$$0 \to \mathrm{Ext}^1_{\mathbb{Z}}(\mathbb{Q}_p/\mathbb{Z}_p, \pi_i\mathrm{L}_n\mathfrak{gl}_1(R)) \to \pi_i\mathrm{L}_{K(1)\vee\cdots\vee K(n)}\mathfrak{gl}_1(R) \to \mathrm{Map}_{\mathbb{Z}}(\mathbb{Q}_p/\mathbb{Z}_p, \pi_{i-1}\mathrm{L}_n\mathfrak{gl}_1(R)) \to 0$$

The first term is precisely the derived p-completion, so it agrees with $\pi_i(\mathrm{L}_n\mathfrak{gl}_1(R))$ for $i \geqslant n+2$. The final term vanishes for $i \geqslant n+3$. It follows that $\pi_i\mathrm{L}_n\mathfrak{gl}_1(R) \to \pi_i\mathrm{L}_{K(1)\vee\cdots\vee K(n)}\mathfrak{gl}_1(R)$ is an isomorphism for $n \geqslant n+3$ and is injective for $n \geqslant n+2$. This implies $\pi_i(F'')$ vanishes for $i \geqslant n+2$. □

Corollary 4.8. *Let R be a $K(1)$-local p-complete $\mathbb{E}_\infty$ ring, and let $\mathbb{G}_m(R)$ be the space $\mathrm{Map}(\mathbb{Z}, \mathfrak{gl}_1(R))$. Then there is a fiber sequence*

$$\mathbb{G}_m(R) \longrightarrow \mathrm{GL}_1(R) \longrightarrow \Omega^\infty R$$

Moreover, $\pi_i(\mathbb{G}_m(R)) = 0$ for $i \geqslant 3$ and $\pi_2(\mathbb{G}_m(R))$ is torsion-free.

Proof. We have the (homotopy) fiber sequence of $\mathbb{E}_\infty$-rings

$$\mathrm{Map}_{\mathbb{E}_\infty}(\mathbb{S}[\pm q], R) \longrightarrow \mathrm{Map}_{\mathbb{E}_\infty}(\mathbb{S}\{\pm q\}, R) \longrightarrow \mathrm{Map}_{\mathbb{E}_\infty}(\mathbb{S}\{q\}, R)$$

which is obtained by quotient q. The left term is equivalent to $\mathbb{G}_m(R)$ by definition, so we obtain the above fiber sequence. Then the truncated and torsion-free arguments follow immediately from Lemma 4.7. □

Now we can finish the proof of Lemma 4.4.

Proof. (Proof of Lemma 4.4) By Lichtenbaum-Quillen conjecture, the map $\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \to \mathrm{L}_{K(1)}\mathrm{TC}(\mathbb{Z}_p[\zeta_p])$ is an equivalence in degree ≥ 2. Therefore, we have the following pullback square

$$\begin{array}{ccc} \mathrm{TC}(\mathbb{Z}_p[\zeta_p]) & \longrightarrow & \tau_{\geqslant 0}\mathrm{L}_{K(1)}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \\ \downarrow & & \downarrow \\ \tau_{\leqslant 0}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) & \longrightarrow & \tau_{[0,1]}\mathrm{L}_{K(1)}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \end{array}$$

By Lemma 4.6, we have shown that Lemma 4.4 is true after replacing $\mathrm{TC}(\mathbb{Z}_p[\zeta_p])$ by $\tau_{\leqslant 0}\mathrm{TC}(\mathbb{Z}_p[\zeta_p])$. Thus it suffices to show that we have a commutative diagram

$$\begin{array}{ccc} \mathbb{S}[S^1] & \longrightarrow & j_{p,0} \\ \downarrow & & \downarrow \\ \mathrm{TC}(\mathbb{S}[q^{\pm 1/p}]) & \longrightarrow & \mathrm{L}_{K(1)}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \end{array}$$

For simplicity we denote $R = \mathrm{L}_{K(1)}\mathrm{TC}(\mathbb{Z}_p[\zeta_p])$. The counterclockwise and clockwise determine two elements in $\mathrm{Map}_{\mathbb{E}_\infty}(\mathbb{S}[S^1], R) \simeq \mathrm{Map}(\Sigma\mathbb{Z}, \mathfrak{gl}_1(R))$. We need to show that they determine the same element in $\pi_0\mathrm{Map}(\Sigma\mathbb{Z}, \mathfrak{gl}_1(R)) \cong \pi_1(\mathbb{G}_m(R))$. We consider the fiber sequence

$$\mathbb{G}_m(R) \longrightarrow \mathrm{GL}_1(R) \longrightarrow \Omega^\infty R$$

By Lemma 4.6, we have shown that both elements are the same in $\pi_1(\mathrm{GL}_1(R))$, so it suffices to prove that $\pi_1(\mathbb{G}_m(R)) \to \pi_1(\mathrm{GL}_1(R))$ is injective. We consider the long exact sequence

$$0 \to \pi_3\mathrm{GL}_1(R) \to \pi_3 R \to \pi_2\mathbb{G}_m(R) \to \pi_2\mathrm{GL}_1(R) \to \pi_2 R \to \pi_1\mathbb{G}_m(R) \to \pi_1\mathrm{GL}_1(R)$$

where $\pi_3\mathbb{G}_m(R)$ equals to 0 by Corollary 4.8. By the computation of [HM+95], we have a splitting

$$R = \mathrm{L}_{K(1)}\mathrm{TC}(\mathbb{Z}_p[\zeta_p]) \simeq \mathrm{J}_p \oplus \Sigma\mathrm{J}_p \oplus \Sigma\mathrm{KU}_p^{\oplus(p-1)}$$

so we obtain $\pi_2\mathrm{GL}_1(R) \cong \pi_2(R) \cong \mathbb{Z}/p$. In order to show that $\pi_1(\mathbb{G}_m(R)) \to \pi_1(\mathrm{GL}_1(R))$ is injective, it suffices to prove $\pi_2\mathrm{GL}_1(R) \to \pi_2(R)$ is an isomorphism. Now since both terms are equivalent to $\mathbb{Z}/p$, it suffices to show it is injective.

Now we need to show that $\pi_2\mathbb{G}_m(R) = 0$. Since $\pi_2\mathbb{G}_m(R)$ is torsion-free, it now suffices to show that $\pi_2\mathbb{G}_m(R) \to \pi_2\mathrm{GL}_1(R)$ is injective, which is equivalent to say that the injection $\pi_3\mathrm{GL}_1(R) \to \pi_3 R$ is an isomorhism. Again by the splitting equation above, we have $\pi_3\mathrm{GL}_1(R) \cong \pi_3(R) \cong \mathbb{Z}/2p \oplus \mathbb{Z}_p^{\oplus(p-1)}$. On torsion part, the cokernal is zero because $\pi_2\mathbb{G}_m(R)$ is trosion-free. On torsion-free part the cokernal is also zero because of the injectivity. We are done.

□

4.2. **The equivalence $j_p \simeq \mathrm{THH}(\mathbb{Z}_p)$.** We now explain the equivalence $j_p \simeq \mathrm{THH}(\mathbb{Z}_p)$. We recall that for a fixed $p > 2$, j_p is defined as $\tau_{\geqslant 0}(\mathrm{KU}_p^{h\mathbb{Z}_p^\times})$.

We first prove the following consequence, which is used as Theorem 3.3.

Theorem 4.9. *There is a commutative square of cyclotomic $\mathbb{E}_\infty$-rings*

$$\begin{array}{ccc} j_p & \longrightarrow & \mathrm{THH}(\mathbb{Z}_p) \\ \downarrow & & \downarrow \\ \mathbb{Z}_p & \longrightarrow & \mathrm{THH}(\mathbb{F}_p) \end{array}$$

Moreover, after applying $\tau_{\geqslant 0}(-)^{t\mathbb{Z}/p}$, we obtain an equivalence $\tau_{\geqslant 0}(j_p)^{t\mathbb{Z}/p} \simeq \mathrm{THH}(\mathbb{Z}_p)$.

The map $j_p \to \mathrm{THH}(\mathbb{Z}_p)$ is constructed as follows. We will see that this map is unique.

Lemma 4.10 ([DR25]; Lemma 1.3.1)**.** *Let $0 \leqslant n \leqslant \infty$ and let A be a p-complete $\mathbb{E}_n$-ring such that the canonical map $\pi_i(A) \to \pi_i(\mathrm{L}_{K(1)}A)$ is an isomorphism for $i \geqslant 2$. Then there is a unique $\mathbb{E}_n$-map $j_p \to A$.*

Proof. By connectivity, it is equivalent to show that there is a unique map $j_p \to \tau_{\geqslant 0}(A)$. By hypothesis, there is a pullback square of $\mathbb{E}_n$-rings.

$$\begin{array}{ccc} \tau_{\geqslant 0}(A) & \longrightarrow & \tau_{\geqslant 0}(\mathrm{L}_{K(1)}A) \\ \downarrow & \lrcorner & \downarrow \\ \tau_{\leqslant 1}\tau_{\geqslant 0}(A) & \longrightarrow & \tau_{\leqslant 1}\tau_{\geqslant 0}(\mathrm{L}_{K(1)}A) \end{array}$$

It now suffices to show that there are unique maps from j_p to the other three terms $\tau_{\geqslant 0}(\mathrm{L}_{K(1)}A)$, $\tau_{\leqslant 1}\tau_{\geqslant 0}(A)$ and $\tau_{\leqslant 1}\tau_{\geqslant 0}(\mathrm{L}_{K(1)}A)$.

First, we note that a map $j_p \to \tau_{\geqslant 0}(\mathrm{L}_{K(1)}A)$ is equivalent to a map $j_p \to \mathrm{L}_{K(1)}A$. This map is unique because the unit $\mathbb{S} \to j_p$ is $K(1)$-equivalence. The uniqueness of other two maps is from a generalized fact. For any 1-truncated, connective and p-complete $\mathbb{E}_n$-ring R, there is a unique map $j_p \to R$ since the unit map $\mathbb{S}_p \to j_p$ is an equivalence after 1-truncation. □

Corollary 4.11. *There is a unique cyclotomic $\mathbb{E}_\infty$-map $j_p \to \mathrm{THH}(\mathbb{Z}_p)$.*

Proof. It is equivalent to show that there is a unique $\mathbb{E}_\infty$-map $j_p \to \mathrm{TC}(\mathbb{Z}_p)$. This follows from Lemma 4.10 and Lichtenbaum-Quillen conjecture for $\mathrm{TC}(\mathbb{Z}_p)$. □

To prove the theorem, we need to compute the homotopy groups of $j_p^{t\mathbb{Z}/p}$. We begin with a well-known result. Recall that for a fixed $p > 2$, ℓ_p denotes the connective p-complete Adams summand, which is equivalent to $\mathrm{ku}_p^{h\mathbb{F}_p^\times}$.

Lemma 4.12. *We have $\pi_*(\ell_p^{t\mathbb{Z}/p}) \cong \mathbb{Z}_p[u^{\pm 1}]$ with u a generator in degree 2.*

Lemma 4.13. *There are isomorphisms*

$$\pi_n(j_p^{t\mathbb{Z}/p}) = \begin{cases} \mathbb{Z}_p & n = 0 \\ \mathbb{Z}_p/k & n = 2k-1 \text{ for } k \in \mathbb{Z} \\ 0 & \text{else} \end{cases}$$

$$\pi_n(j_p^{t\mathbb{Z}/p}/p) = \begin{cases} \mathbb{F}_p & n = 2pm \text{ or } 2pm-1 \text{ for } m \in \mathbb{Z} \\ 0 & \text{else} \end{cases}$$

Moreover, the projection map $j_p^{t\mathbb{Z}/p} \to \mathbb{Z}_p^{t\mathbb{Z}/p}$ induces isomorphisms on even homotopy groups.

Proof. We consider the projection $j_p \to \mathbb{Z}_p$ and $\ell_p \to \mathbb{Z}_p$. Since $\mathbb{Z}_p^\times$ acts trivially on $\pi_0(j_p) = \pi_0(\ell_p) = \mathbb{Z}_p$, we have a pullback diagram, where the horizontal arrows are given by comparison maps of homotopy limits.

$$\begin{array}{ccc} j_p & \longrightarrow & \ell_p^{h\mathbb{Z}} \\ \downarrow & \lrcorner & \downarrow \\ \mathbb{Z}_p & \longrightarrow & \mathbb{Z}_p^{h\mathbb{Z}} \end{array}$$

We can apply $(-)^{t\mathbb{Z}/p}$ to the diagram and using the fact that $(-)^{h\mathbb{Z}}$ is a finite limit, we obtain the following pullback diagram.

$$\begin{array}{ccc} j_p^{t\mathbb{Z}/p} & \longrightarrow & (\ell_p^{t\mathbb{Z}/p})^{h\mathbb{Z}} \\ \downarrow & \lrcorner & \downarrow \\ \mathbb{Z}_p^{t\mathbb{Z}/p} & \longrightarrow & (\mathbb{Z}_p^{t\mathbb{Z}/p})^{h\mathbb{Z}} \end{array}$$

With Lemma 4.12 in hand, one can compute that

$$\pi_n((\ell_p^{t\mathbb{Z}/p})^{h\mathbb{Z}}) = \begin{cases} \mathbb{Z}_p & n = 0 \\ \mathbb{Z}_p/pk & n = 2k-1 \text{ for } k \in \mathbb{Z} \\ 0 & \text{else} \end{cases}$$

The pullback diagram will result in a long exact sequence, by analyzing which we obtain the computation of $j_p^{t\mathbb{Z}/p}$ consequently.

□

We now give Theorem 4.9 a sketch of proof. The map $j_p \to \mathrm{THH}(\mathbb{Z}_p)$ is constructed uniquely and it fits into the following diagram automatically.

$$\begin{array}{ccc} j_p & \longrightarrow & \mathrm{THH}(\mathbb{Z}_p) \\ \downarrow & & \downarrow \\ \mathbb{Z}_p & \longrightarrow & \mathrm{THH}(\mathbb{F}_p) \end{array}$$

We need to show that after applying $(-)^{t\mathbb{Z}/p}$ and taking the connective part, we obtain an equivalence $\tau_{\geqslant 0}(j_p)^{t\mathbb{Z}/p} \simeq \mathrm{THH}(\mathbb{Z}_p) \simeq \mathrm{THH}(\mathbb{Z}_p)^{t\mathbb{Z}/p}$, while the second equivalence is obtained by Segal conjecture. To see the first equivalence, we analyze everything after modulo p. By Lemma 4.13, $j_p^{t\mathbb{Z}/p}/p \to \mathbb{Z}_p^{t\mathbb{Z}/p}/p$ induces isomorphisms on even homotopy groups. The same is true for $\mathrm{THH}(\mathbb{Z}_p)/p \to \mathrm{THH}(\mathbb{F}_p)/p$ and $\mathbb{Z}_p^{t\mathbb{Z}/p} \to \mathrm{THH}(\mathbb{F}_p)$ by [NS18]. Therefore, by the methods in Lemma 4.3, $\pi_*(j_p^{t\mathbb{Z}/p}/p) \to \pi_*(\mathrm{THH}(\mathbb{Z}_p)/p)$ are isomorphisms on even degrees and degree 1. Both sides admit ring structures which are determined by generators in even degrees and degree 1.

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